

Student-Faculty Rapport

Today's Objective

After reviewing the figures and tables:

- Find one that you really liked, take some notes as to what you liked.
- Find one that left you wanting or that wasn't clear, take some notes as to why this might be.
- Discuss that with your partner for the day.

Tomorrow's Objective (not to be submitted)

Pick at least one of these to read, and then consider:

- What you feel helps you build rapport with your faculty peers?
- What do you see in your mentors that works well for you?
- Where might you be able to advocate for your own needs to support your belonging and success with your mentors?

As you start your career, or join a research lab, or start a collaboration with others, I want you to empower yourself to build the rapport and relationships you need to support your collective success.

#ASEEVC



JUNE 22 - 26, 2020

ASEE'S VIRTUAL CONFERENCE

At Home with Engineering Education

CONFERENCE PROGRAM

PRESENTED BY UNIVERSITY OF MARYLAND

 AMERICAN SOCIETY FOR
ENGINEERING EDUCATION

Kara L. Hjelmstad, Arizona State University
Prof. Keith D. Hjelmstad, Arizona State University
Lydia Ross, Arizona State University
Prof. Robert J. Culbertson
Prof. James A. Middleton, Arizona State University
Dr. Claire Fletcher Honeycutt, Arizona State University
Ke Liu, Arizona State University

Faculty Development Mini-modules on Evidence-based Inclusive Teaching and Mentoring Practices in Engineering

Dr. Sarah Ilkhanipour Rooney, University of Delaware
Prof. Joshua A. Enszer, University of Delaware
Dr. Julia A. Maresca, University of Delaware
Dr. S. Ismat Shah, University of Delaware
Prof. Sheldon Allister Hewlett, University of Delaware
Prof. Jenni M. Buckley, University of Delaware

International Faculty Professional Development: Utilizing Hybrid Environments to Deepen Learning and Grow Community

Dr. Mary Slowinski, College of the Canyons
Mrs. Gabrielle P. Temple
Dr. Kenneth Walz, Madison Area Technical College

T257B - Faculty Development Research

10:00 A.M. - 10:20 A.M., ONLINE, A VIRTUAL CONFERENCE

Sponsor: Faculty Development Division

**Moderators: Sunay Palsole, Texas A&M University;
Stephanie Cutler, Pennsylvania State University**

What's new in faculty development research? Come find out!

Extending Faculty Development through a Sustainable Community of Practice

Sarah Hoyt, Arizona State University
Dr. Lindy Hamilton Mayled, Arizona State University
Prof. Stephen J. Krause, Arizona State University
Prof. Keith D. Hjelmstad, Arizona State University
Kara L. Hjelmstad, Arizona State University
Dr. Claire Fletcher Honeycutt, Arizona State University
Dr. Eugene Judson, Arizona State University
Lydia Ross, Arizona State University
Prof. James A. Middleton, Arizona State University
Prof. Robert J. Culbertson, Arizona State University
Ke Liu, Arizona State University

Improving Automated Group Assignments in an Academic Setting

Prof. Petra Bonfert-Taylor, Dartmouth College
Mr. Christopher Miller, Dartmouth College

Optimizing Student-Faculty Rapport for the Engineering Classrooms: Dimensioning the Behaviors that Matter

Dr. Fethiye Ozis P.E., Northern Arizona University
Dr. Kyle Nathan Winfree, Northern Arizona University

The Role of Teaching Self-Efficacy in Electrical and Computer Engineering Faculty Teaching Satisfaction

Mr. Kent A. Crick, Iowa State University
Elise A. Frickey, Iowa State University
Dr. Lisa M. Larson Ph.D., Iowa State University of Science and Technology
Prof. Mack Shelley, Iowa State University of Science and Technology

T271 - NSF Grantees: Diversity 1

10:00 A.M. - 10:20 A.M., ONLINE, A VIRTUAL CONFERENCE

Sponsor: NSF Grantees Session

Moderator: Doris Espiritu, Wilbur Wright College- One of the City Colleges of Chicago

Presentations from groups with current NSF-funded projects focused on understanding and supporting diverse populations within engineering.

(Mis)match of Students' Country of Origin and the Impact of Collaborative Learning in Computer Science

Prof. Nicholas A. Bowman, University of Iowa
Lindsay Jarratt, University of Iowa
Dr. KC Culver, University of Southern California
Dr. Alberto Segre, University of Iowa

Recruiting More U.S. Women into Engineering Based on Stories from Morocco

Prof. Rani W. Sullivan, Mississippi State University
Miss Soundouss Sassi, Mississippi State University
Dr. Mahnas Jean Mohammadi-Aragh, Mississippi State University

Does Stereotype Threat Affect Creative Thinking in Female Engineering Students? A Behavioral and Neurocognitive Study

Dr. Rafal Jonczyk, Adam Mickiewicz University, Poland, and Pennsylvania State University
Ms. Yushuang Liu, Pennsylvania State University
Dr. Danielle S. Dickson, Pennsylvania State University
Dr. Gül E. Okudan-Kremer, Iowa State University of Science and Technology
Prof. Zahed Siddique, University of Oklahoma
Prof. Janet van Hell, Pennsylvania State University

Reimagining Energy Year 2: Integrating CSPs into Course Development

Optimizing Student-Faculty Rapport for the Engineering Classrooms: Dimensioning the Behaviors That Matter

This research paper describes the findings from an exploratory study. Student retention in engineering disciplines, from program initiation through commencement, is recognized as a challenge by higher learning institutions across the US. Numerous studies have identified that professors who can establish strong and positive rapport with their students have an immediate and positive impact on students' learning, engagement, motivation and academic success, resulting in a positive long-term influence on retention. Previous work has defined fifteen specific faculty behaviors that establish positive rapport between students and professors in other disciplines. However, these past studies may not be generalizable for engineering, and fall short of identifying the differences across student academic maturity and engineering discipline. Given the potential for strong impact on engineering students' experiences, and faculty resource limitations, we sought to elucidate if some behaviors have a higher potential for impact than others, and if so, which faculty behaviors may best contribute to building faculty-student rapport. With these insights, engineering faculty can be trained to selectively focus on those which are expected to have the greatest return on investment. To support the primary aim, students from multiple disciplines in an engineering college were surveyed and asked to rank their perceptions of faculty behaviors that best establish rapport with engineering students. The survey asked students to self identify their gender. This was used to identify how these factors may influence ranking of rapport supportive behaviors. Additionally, this survey asked students both their academic program (the discipline within engineering) and their degree progression, in order to identify how these factors impact such rankings.

Based on findings in the literature, and given the scope of this project, the authors anticipate that the most effective way to broaden students' retention in engineering education is through establishing rapport between engineering professors and their students. Using the results of this study, we can design interventions aimed at faculty member's ability to establish positive rapport, which in turn creates a respectful environment, and removes the barriers to cultivation of diverse student retention in engineering disciplines.

1: Introduction

According to the U.S. Bureau of Labor Statistics, STEM occupations are expected to grow by 8.8% from 9,708.3 (2018) to 10,566.8 (2028). This is a significant growth when compared to the percent growth for all occupations (5.2%). There had been warning signs and continuing concern about the shortage of qualified candidates from a diverse pool to fill these jobs. In this paper we discuss retention in science and engineering, by focusing on specific predictors of persistence including gender, student- instructor rapport, and year of study [1].

1.1: Retention in Science and Engineering

In a 2016 study, nearly half (45%) of all high school seniors indicated an intent to study science and engineering (S&E), yet in the 2015 survey of full-time undergraduates, just more than one third (37%) of undergraduate enrollments were in S&E programs, indicating there exists a disconnect between enrollment and graduation rates. In 2015, out of nearly two-million bachelor's degrees earned; less than one-hundred thousand were in engineering (5.2%) (NSB Appendix Table 2-21 [2]). "We are graduating fewer engineers now than 20 years ago, both in terms of absolute numbers and as a percentage of all college degrees" [3]. This is also reflected in the National Science Board (NSB) cohort study which identified that more than one in six (16.3%) of students who declared their major as S&E at enrollment, were no longer enrolled in any institution three years after their initial college enrollment [2]. These numbers indicate that students lose enthusiasm or interest in S&E somewhere between high school, when many are interested in an S&E career, and graduation. Further, students are aware of this, as nearly half (49%) of surveyed engineering juniors report having doubts about majoring in engineering. Retaining students until graduation is recognized as a challenge by higher learning institutions across the US [4]. Due to the heavy focus on content, engineering in college turns into a "closed club" and becomes less attractive for students to persist in [3].

1.2: Predictors of Persistence

Seymour and Hewitt [5], in a landmark study, stated the concerns about highly qualified students' leaving S&E at about 50% rate, and that ratio heightened at a greater disproportion for women and students of color, despite a serious national effort to improve their recruitment and retention.

Persistence in science and engineering programs can however be partially explained by a known set of variables. For example, learning and course success, directly associated with grades and GPA, are identified as one of the most important predictors of retention [4]. Here we explore two other factors, which we hypothesize have an interaction effect.

1.2.1: Gender

A student's gender has also been identified as a predictor of success in S&E programs [3]. Engineering, excluding computer science, is showing a declining representation of women in contrast to other STEM majors [6]. Although women have nearly the same grades as their male counterparts, women leave engineering in greater percentages than men, often going on to complete other demanding majors [7]. Factors that impact women and other underrepresented minority students' decisions to leave include several academic, social and career issues including, but not limited to, dealing with prejudice, belongingness, lack of academic preparedness, feeling overwhelmed by the pace, poor teaching and being treated differently by faculty [5], [8], [9].

1.2.2: Rapport

One factor not commonly considered, is student-instructor rapport. A student's positive perception of student-instructor rapport is arguably one of the most under-considered factors identified to positively impact retention. Rapport is defined as the feeling of mutual trust and understanding developed between two individuals through frequent interpersonal and enjoyable interactions across different settings [10] - [13]. When it is applied to a learning environment, whether in a presentation, workshop, or a lecture, "being able to establish rapport" is one of the seven essential abilities of an effective presenter, whom then "opens the door and invites learning" [14].

Numerous studies have examined student-professor rapport in non- science and engineering undergraduate programs, and its impact on various student outcomes including student learning, engagement, motivation, grades [4], [15] - [20], and attentiveness and studying efforts for the course [9], [15], [19]. Students were less likely to miss class when they experienced rapport with their professors [16]. It is safe to argue that class attendance as a critical component of learning and success [21]; further, it is known that it can be promoted by establishing rapport with students [16], [19]. Student-instructor rapport is also related to, and a predictor of, expected and actual grades [15], [16]. By establishing rapport, professors convey confidence in their students' ability to do well in the course [16]. This student-centered and growth-mind set approach engraved in rapport translate into greater intrinsic motivation and therefore higher student achievement [22], [16] .

Students also recognize the value of rapport, as they report rapport to be a key component of an effective professor [16], [24], [25]. This component can be improved by being encouraging, getting to know students, and maintaining frequent contact [15].

Knowing that rapport contributes to student attendance, learning, grades, intrinsic motivation, engagement, and degree completion, and that rapport can be modified through instructor behaviors, we sought to identify the behaviors most relevant to students in the science and engineering disciplines.

1.3: Hypothesis

Based on these collective findings, we sought to explore the faculty behaviors that have the highest potential for impact on student-instructor rapport. Further, we hypothesize that, within the science and engineering disciplines, there may exist differences in the perception of instructor behaviors that influence rapport depending on a) program of study, b) year of study, and c) gender. Therefore, the specific goals of this research project are to answer the following research questions: 1) What are the top quartile behaviors that establish positive rapport between engineering professors and engineering students; 2) Are there preferential differences for faculty behaviors between a) different engineering programs, b) the different years of student progression, or c) male and female students?

Existing studies in the engineering classroom noted the importance of faculty interactions but fall short of identifying if there are differences in level of rapport across grade levels. To the best of our knowledge, this work is the first in the field to specifically look at multiple engineering programs to identify the level of rapport across grade level, and to explicitly look at instructors' rapport building behaviors with an intention to develop a faculty training. We will broaden our impact by training engineering faculty on best practices that promote rapport between students and professors to tackle the difficulties of students' professional formation of engineering identity, first at our local institution, and later through publication of key findings from those efforts. This paper presents the results of the investigation focused on the student-instructor rapport building behaviors.

2: Methods

Students from multiple disciplines in an engineering college were surveyed online and asked to rank their perceptions of faculty behaviors that best establish rapport with themselves. The participating students were recruited using a college-wide all-student Listserv, and were reminded of the survey two weeks after the original notification. The survey asked students to self-identify their program of study, year of study (including graduate student status), and gender (Appendix A). The survey then asked fifteen questions about faculty behaviors that influence student-instructor rapport. These questions were derived from literature, which investigated student-instructor rapport in other disciplines. Participants selected responses using a five-point likert scale that ranged from "Strongly Disagree" to "Strongly Agree" for each statement of "When professors ..."

Relative representation of each program of study, year of study, and gender was calculated and compared with the representative ratios for the university's college of engineering. Response groups that had low representation, less than or equal to two responding participants, were excluded in subgroup analyses.

Responses were then sorted by the descriptors, grouping those responses by program of study, year of study, gender, and combined program-year of study. Overall responses for all students were also calculated, as well as a normalized and unbiased response that controlled for differences in program of study response group sizes. To normalize, the mean response from each of the seven programs were then averaged together, giving equal one-seventh weight to each program of study.

Response categories of "Strongly Disagree," "Disagree," "Neutral," "Agree," and "Strongly Agree" were assigned corresponding ordinal ranks of one through five respectively. The likert value responses were averaged for these groups in order to represent a "general consensus" response. Mean likert responses near one for any given question reflect a strong disagreement that the given behavior is an important factor for student-instructor rapport, while mean responses near five indicate a strong agreement that the behavior is an important factor for student-instructor rapport. Median was not calculated, as the median of five categorical values obscures small differences in response. Likewise, standard deviation was not calculated for these

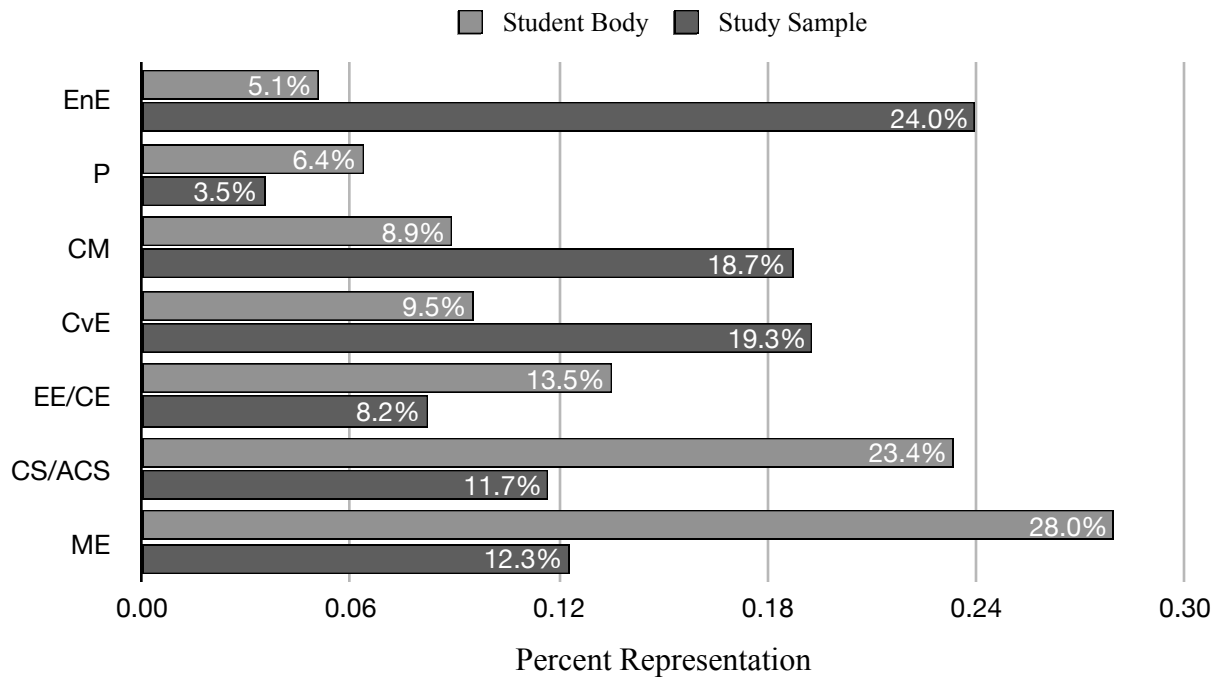


Figure 1: Shown here is the relative representation of students in the surveyed college, and the relative representation of students that responded to the survey. Disciplines represented include Environmental Engineering (EnE), Physics (P), Construction Management (CM), Civil Engineering (CvE), Electrical Engineering and Computer Engineering (EE/CE), Computer Science and Applied Computer Science (CS/ACS), and Mechanical Engineering (ME), and are shown in order of study body representation.

responses, given that standard deviation is not an appropriate mechanism of reflect measure spread for ordinal ranked categorical data. A histogram was created to aid in understanding the skew and spread of responses to each question.

Quartiles were calculated for each question. With limited time and other resources, faculty aren't necessarily able to increase efforts in all behaviors listed. Thus, prioritization of the key factors may be a valuable approach to improving rapport. Questions identifying top quartile response have been selected as areas of faculty behavior that should be considered for such focus.

3: Results

The college of engineering study body was approximately 2600 students in size. From these 2600 students, 171 students responded to the survey. Figure 1 shows the relative percentage of the student body for each program of study, and the percentage of student response from these programs. As can be seen, Mechanical Engineering is the largest study body in this college of engineering, and is represents approximately one of eight of the responses in this study, a representation ratio seen across most of the programs surveyed. Environmental Engineering and Physics are two exceptions to this trend. Environmental Engineering has the smallest body of

student representation, but also had the highest representation in this survey. Physics is the second smallest program of study, and also saw the smallest relative representation in the survey.

In the college, and across disciplines, females represent approximately 19.6% of the student body, 80.4% were male, and less than one percent of the student body have not identified with one of these genders. Response rates differed in that 40.5% of respondents were female, 58.3% were male, and 1.2% of the students preferred not to respond to this question. Freshman, sophomores, juniors, and seniors represent 28.5%, 21%, 19.8%, and 30.7% percent of the undergraduate study body across disciplines. This is similar to that seen in the survey responses, with 13.5%, 30.7%, 29.4%, and 26.4% percent response representation respectively across year of study.

Shown in Figure 2 are the net responses for all questions, across all student respondents. Notably, no questions elicited more than ten “Strongly Disagree” responses and most questions received more “Strongly Agree” responses than other response types. Four questions had more “Agree” than “Strongly Agree” responses, and one had an equal number of “Agree” and “Strongly Agree” responses.

Looking at top quartile responses, the following was found for each group. Groups considered include a) the entire group, b) the entire group and normalized for response rate, c) each program of study, d) year of study, e) gender, and f) combined program and year of study. Questions and a corresponding question keyword are shown in Figure 2.

3.1 The Entire Group

The majority of the responses were in the range of “Strongly Agree”, which tells us that the students found all behaviors to be relevant to student-instructor rapport. The quartile analysis revealed that there are four “top” behaviors that impact student-faculty rapport. In order, these are “When professors are respectful” (respect, mean likert response 4.75), “When professors keep up with his/her email” (email, 4.67), “When professors use relevant class examples” (examples, 4.65), and “When professors are enthusiastic” (enthusiasm, 4.65).

Normalizing for response representation, only three behaviors emerge in the top quartile; “When professors are respectful” (respect, 4.73), “When professors keep up with their email” (email, 4.67), “When professors use relevant examples” (examples, 4.63).

3.2 Between Programs of Study

Top quartile behaviors for each program of study are shown in Table 1. This represents all years of study for each of these programs. Graduate students from the Environmental and Civil Engineering programs responded to this survey and are included in this analysis. An ANOVA was used to evaluate differences in ratings between the different disciplines. This analysis did not reveal a statistical difference across all groups.

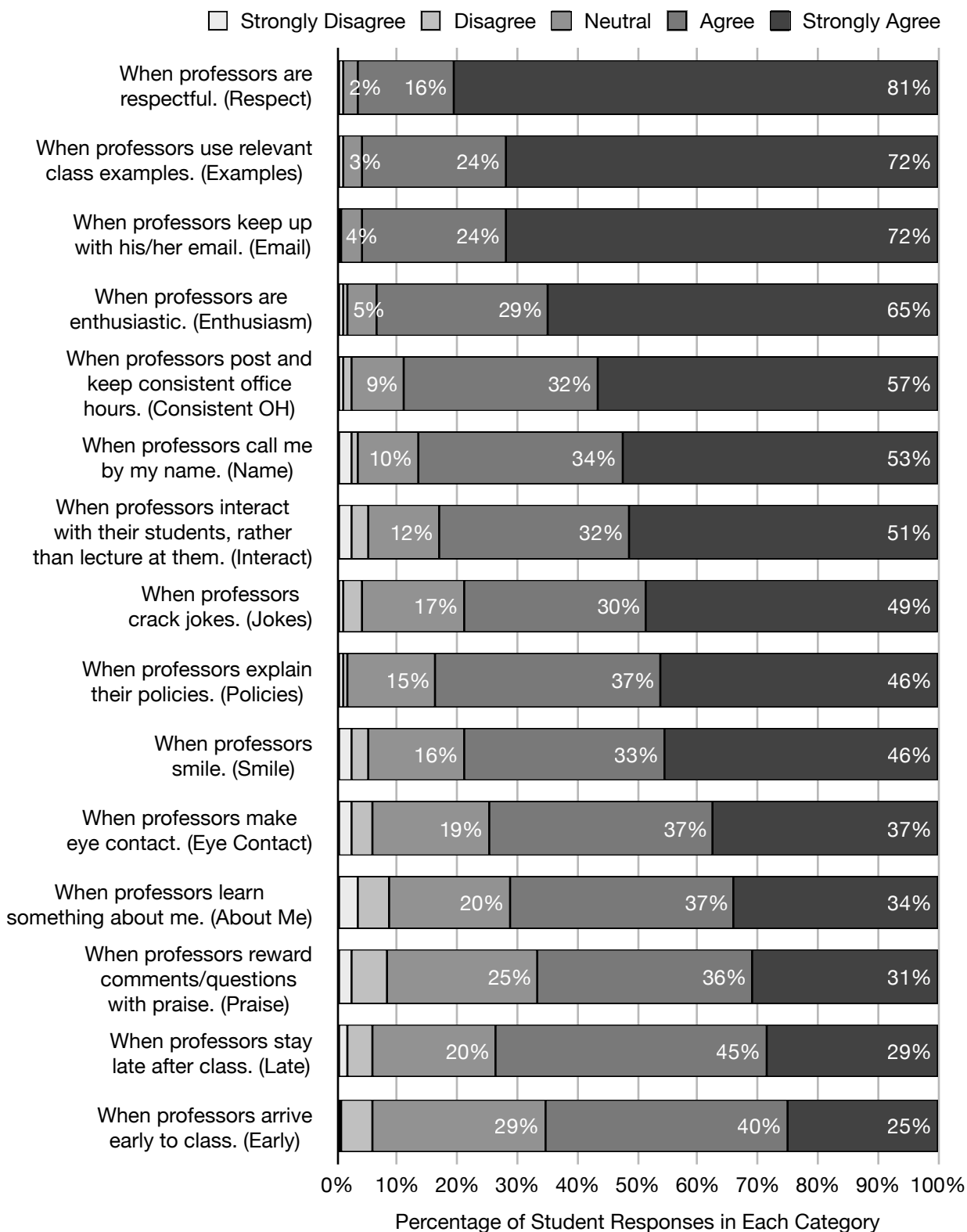


Figure 2: Shown here are the relative responses of “Strongly Disagree” to “Strongly Agree” for each of the questions asked in the survey. This includes all respondent students in raw form across all programs of study, years of study, and genders, and is ordered in decreasing levels of “Strongly Agree.”

Table 1: Top quartile faculty behaviors, as stratified by program of study.

Ranking	EnE	CvE	CM	EE / CE	CS / ACS	ME	P
1	Respect (4.93)	Respect (4.73)	Respect & Email (4.72)	Email (4.71)	Respect (4.75)	Email (4.71)	Respect (4.83)
2	Examples (4.91)	Enthusiasm (4.67)	Examples & Enthusiasm (4.59)	Respect & Examples (4.64)	Email (4.60)	Examples (4.67)	Email, Examples, & Consistent OH (4.67)
3	Enthusiasm (4.78)	Email (4.58)		Policies (4.43)	Consistent OH (4.55)	Respect (4.48)	
4	Email (4.69)	Smile (4.48)			Examples (4.50)	Consistent OH (4.33)	

Respect: "When professors are respectful."

Examples: "When professors use relevant class examples."

Enthusiasm: "When professors are enthusiastic."

Email: "When professors keep up with his/her email."

Smile: "When professors smile."

Policies: "When professor explain their policies."

Consistent OH: "When professors post and keep consistent office hours."

Table 2: Top quartile faculty behaviors, as stratified by year of study.

Ranking	Freshmen	Sophomore	Junior	Senior	Graduate
1	Respect (4.80)	Email (4.88)	Respect (4.67)	Respect (4.77)	Examples (4.88)
2	Name (4.61)	Respect (4.78)	Email (4.66)	Email (4.58)	Respect & Enthusiasm (4.75)
3	Examples (4.58)	Examples (4.76)	Examples & Enthusiasm (4.52)	Examples (4.57)	Email (4.50)
4	Email (4.54)	Consistent OH (4.57)	<i>Consistent OH (4.49)</i>	Consistent OH (4.42)	<i>Interact, Name, Smile, & Consistent OH (4.25)</i>
5+	<i>Interact & Enthusiasm (4.35)</i>	Enthusiasm (4.55)		Enthusiasm (4.36)	

Italic lettering here reflects the top behavior in the second quartile.

Interact: "When professors interact with their students, rather than lecture at them."

The arrows here reflect changes in the student rating of each faculty behavior. The darker, larger arrow reflects the consistently highest rated behavior. The lighter, mid-weight arrow represents the consistently second highest rated behavior. Finally, the lightest, smallest weight arrow represents the third highest rated behavior. Not reflected, but seen in this table, are behaviors which demonstrate varying priority from the students as a function of academic year. For example, being known by name is important for freshman and graduate students, but does not present as a top quartile behavior in sophomore, junior, or senior students.

3.3 Between Years of Study

To answer the question of, if student academic progress status changes the student perception of what faculty behaviors contribute to student-instructor rapport, all students were grouped by year of study. This is shown in Table 2.

Table 3: Top quartile faculty behaviors, as stratified by gender.

Ranking	Female	Male
1	Respect (4.81)	Respect (4.71)
2	Email (4.77)	Email (4.61)
3	Examples (4.74)	Examples (4.59)
4	Enthusiasm (4.59) & Consistent OH (4.59)	Enthusiasm (4.54)

Enthusiasm and Consistent Office Hours were equally ranked by female respondents. All other behaviors were equally rated between female and male students.

3.4 Between Genders

Across all program and years of study, students were grouped by gender to identify gender specific influence in perceptions of faculty behaviors that build rapport. Shown in Table 3, the ranking between genders was exceptionally consistent.

4. Discussion of the Results

As seen above, regardless of how the responses are stratified, some behaviors are consistently identified by the students as determining, top quartile behaviors: “When professors are respectful,” “When professors keep up with his/her email,” and “When professors use relevant class examples.”

4.1 The Entire Group

Whether “Agree” or “Strongly Agree,” these positive responses combined constitute 97-94% of all responses in the top quartile behaviors. The next three quartiles received 89-65% “Agree” and “Strongly Agree” responses. This demonstrates that the top behaviors are clearly and strongly valued by the vast majority of students. Regardless of a student’s exact engineering program of study, instructors looking to build rapport should consider respect, email, relevant examples, and enthusiasm as key factors to leverage.

4.2 Between Programs of Study

As shown in the results and Table 1, there are several interesting differences between the various disciplines. This may reflect discipline culture differences, expectations of the students, or even differences in the instructors of those disciplines. For example, Electrical / Computer Engineering (EE/CE) and Mechanical Engineering (ME) students both rank “Email” higher than “Respect,” while the other disciplines all rank “Respect” as the top behavior. This suggests that students in these disciplines feel communication is more important than reverence in building a

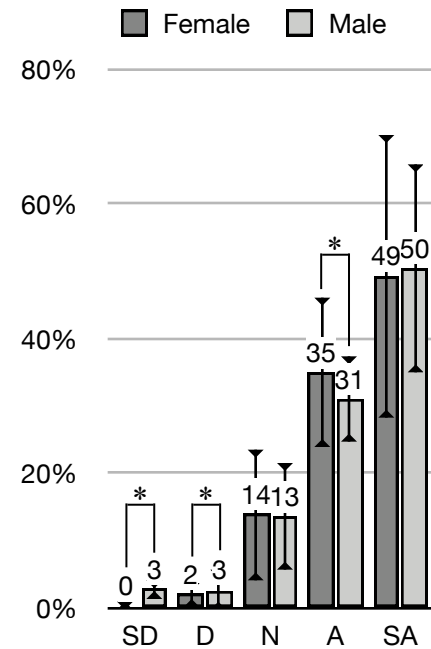


Figure 3: Shown here is the relative distribution of Strongly Disagree (SD) to Strongly Agree (SA) responses across all questions for female and male students. Numbers above each bar reflect the percentage of responses for each of the SD-SA categories. Error bars represent the variance (in standard deviation) of responses for each category and gender. Statistically significant differences are represented with an *.

harmonious student-faculty relationship. It is unclear if this reflects differences in communication of instructors between EE/CE/ME and the other disciplines, or if this may reflect differences in the expectations of the students between these disciplines.

Likewise, “When professors use relevant class examples” was in the top quartile of response for all programs of study except Civil Engineering (CvE). CvE is a discipline that students are exposed to real world examples every day. Assuming then that CvE students capitalize on this exposure, “relevant examples” may be inferred, and thus not a behavior that these students think about directly impacting rapport. Interestingly, CvE students were the only group that listed “Smile” in the top quartile. Yet it is unrealistic to think that CvE instructors smile any more, or less, than other discipline instructors; this may indicate other social or core value differences between CvE and other discipline students.

“When professors post and keep consistent office hours” came up in the top quartile for Computer / Applied Computer Science (CS/ACS), ME, and Physics. These students might prefer out-of-class clarification of abstract or otherwise challenging topics. This may be exemplified with in-class clarification for EE/CE students, whose responses for “Respect” and “Examples” are tied as the second most impactful instructor behaviors. Both suggest that additional learning modalities are important.

Another unique observation is the inclusion of “Policies” in the top quartile of EE/CE student respondents; no other disciplines reported “Policies” as a top quartile behavior. EE/CE students may have higher expectations of the clarity of “the class rules.”

Finally, “Enthusiasm” appeared in the first quartile for EnE, CvE, and CM respondents. These three disciplines are keenly related in the college of this study, suggesting that faculty may be exceptionally enthusiastic, enough so such that students recognize and appreciate such efforts.

4.3 Between Years of Study

It was interesting to notice that Freshmen uniquely lists “call by name” in their top quartile. These students, who recently started college, quite possibly feel like a number in a crowd of unfamiliar faces and thus realize the importance of their professors calling them by names more than other groups. Both sophomores and seniors listed “keeping consistent office hours” in their top quartile. One has to wonder if this suggests that sophomores are starting to recognize the value of meeting with their instructors. Likewise, seniors might recognize the significance of seeking help to ensure class success - most students have very few contingency options should they fail a required class in their last year or semester. Juniors listed enthusiasm as an important factor to establish rapport between themselves and their professors. Enthusiasm may support students through a “junior year slump.”

Most instructors might express the desire to address all of these program year specifics, across all phases of student progression, but these findings suggest that targeting specific behaviors might be a more efficient and effective approach. In short, Freshmen benefit more from being known

by name. Consistent office hours is key for students just starting into the harder topics of their program (generally sophomores) and for students who recognize the importance of finishing a class successfully to graduate (seniors). Juniors appreciate the contagious nature of the enthusiasm to infuse into their day in day out routine.

4.4 Between Genders

Females are, on average more biased than males, to agree that the behaviors listed support rapport building. As shown in Figure 3, this is demonstrated with the statistically significant difference in Agree (A) rates between the genders. The other side of this difference is reflected in the statistically significant difference in Strongly Disagree (SD) and Disagree (D) responses. Females provided a generally stronger agreement to the statements than males.

Both females and males listed “when professors are respectful” as their top choice among all the behaviors that establish rapport. They were in fact consistent for the next three choices as well for both genders. Within most behaviors, there was no gender difference between the behavior ranking, as shown in Table 3. Figure 3 shows that females provided a stronger agreement across all behaviors; while the rate of Strongly Agree (SA) is not statistically significantly different between genders, there does exist a statistical difference between genders in the rate of Agree (A) responses. This positive bias is mirrored in the tied mean response of the fourth ranked behaviors (Table 3) in the top quartile, the response of females that “keeping consistent office hours” is a top behavior in establishing rapport, possibly indicating that female students consider both written and in-person communication important in establishing rapport.

5. Conclusion

Respect for the students was first or second place behavior in all approaches of data stratification. Across program of study, and year of study, and gender, this behavior was clearly and consistently identified as important. Freshman stand out with the unique ranking of being known by name. In addition, females appreciate posted and consistently held office hours.

Our ultimate goal is to pave a path to engineering for more students from a broad range of backgrounds, interests, and experiences by increasing rapport between students and instructors. The results of this work reveal the best practices that particularly help students to establish student-instructor rapport. The best practices will be used to inform all engineering faculty at our own institution, so that they can strategically establish rapport in their classrooms. Establishing rapport in engineering classrooms will improve students’ attendance, learning, grades, intrinsic motivation, and engagement. We expect this is especially important for our underrepresented minority student groups. With the help of the training, we anticipate that establishing rapport will retain more minority students in engineering fields. Once the representation increases, it will be easier than ever to find role models for engineering students from various backgrounds, positively influencing lack of belongingness issues. It will be a positive feedback cycle of rapport creating student learning, success, motivation, and retention, and therefore increased representation in the field.

Using the results of this study, we can design interventions aimed at faculty member's ability to establish positive rapport, which in turn creates a respectful environment, and removes the barriers to cultivation of diverse student retention in engineering disciplines. The findings of this project could be disseminated to the engineering faculty through a blended training, online training coupled with a half day in-person workshop. This blended training format has been developed by Tingerthal and Ozis to model effective practices given the 'expert' nature of the participants and the limited time available, further information on this training format can be found in their paper [26]. The participants are expected to engage in two to three hours of online self-study, by reading the selected literature on rapport and professor behaviors, prior to a four-hour in-person structured seminar. The advantage of this format coupled with facilitators' experience could be used to engage in higher level discussions on the findings of our study. The half-day in-person seminar is structured to provide activities that would reinforce the topics explored within the online modules and would provide a discussion forum and networking opportunity for new instructors. The details about the activities are available upon request.

6. Challenge and Limitations

The challenges experienced during this work were related to the data collection, and the small sample size. One should be quick to note though, that it is known that all data based studies are constrained with the quality and quantity of collected data, and generalization beyond the context of the original study may violate context specific assumptions. The survey, as a Google form, was sent via email by using listservs, and the response rate was relatively low for some sub groups. A second reminder was sent to specific targeted groups as a follow up, after the first two weeks still had low correspondence.

Limitations to this study included that the Graduate students were only from Civil and Environmental Engineering Graduate program, therefore results may not be applicable for the graduate student body at large. Among undergraduate respondents, the largest portion (26.3%) were from Environmental Engineering Program. Also, our college is unique to have Applied Sciences combined with Engineering, we had 3.5% of the respondents from Physics.

7. Future Work

We plan to continue with additional surveys, and seek responses from broader student body to determine and rank the behaviors establishing rapport in engineering classrooms. In addition, our goal is to further develop the blended faculty professional development and educate our faculty on the behaviors that establish rapport. To understand the real impact on students, we will continue to study the instructor behavior, and student feedback correlating the impact of rapport on student resilience and retention in a longitudinal study.

It is important for us to hear students' perceptions as well. A future study will seek to understand how our students define "respect" as compared to how faculty define it.

References

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To Infinity and Beyond: Boosting URM Students' Career Trajectories Through Professional Experiences

Dr. Fethiye Ozis P.E., Northern Arizona University

Dr. Fethiye "Faith" Ozis is a senior lecturer in the civil and environmental engineering department at Northern Arizona University. Dr. Ozis holds a B.S. in environmental engineering from the Middle East Technical University, Ankara, Turkey and a Ph.D. from the University of Southern California, Los Angeles. She is a licensed Professional Engineer, Environmental, in Arizona. Dr. Ozis enjoys every dimension of being an engineering educator. She conducts research related to engineering classrooms and innovative pedagogical strategies. Her own intersectionality led to her passion in promoting and researching pathways into STEM especially for underrepresented minority groups.

Dr. Kyle Nathan Winfree, Northern Arizona University

Dr. Winfree is the Associate Director for Undergraduate Programs in the School of Informatics, Computing, and Cyber Systems at Northern Arizona University. His research focuses on wearable technologies as applied to health assessment and rehabilitation. He teaches in Computer Science, Electrical Engineering, and Informatics.

Ms. Elizabeth Glass, Northern Arizona University

Elizabeth has been engaged with career development programming for STEM majors as well as at NAU since 2012. Before that, her own career path has been dotted with many exciting and rewarding ventures, some of which include 7 years as a high school science teacher (2 of which were at an international school in Mongolia), and 6 years in logistics and other science support for NSF grantees in the Antarctic.

To Infinity and Beyond: Boosting URM Students' Career Trajectories Through Professional Experiences

Abstract

It is hard to deny the impact of experiential learning through internships on engineering education. Likewise, students may also benefit from professional experiences that are not traditional internships. From shadowing, to full-blown hands-on internships, experiential learning provides students with the ability to see application of theory beyond classrooms, to learn to efficiently practice their art, develop transferable skills, and add further value to their academic career.

Engineering educators, and engineering industry leaders, have long recognized the value of and the need for “practice ready” graduates. Today, as the world grapples with the work-from-home and social distancing guidelines necessary to slow the spread of COVID-19; we are in need of “practice ready” graduates more than ever before. The degree to which engineering graduates are prepared to perform on the job can be further improved by establishing strong and effective college-industry partnerships that develop meaningful and diverse professional experiences. The value of these experiences, resultant of strong college-industry partnerships, include but are not limited to: refining and expanding students’ professional identity, practice readiness, and improving resilience.

In this paper, we considered professional experiences from the students’ perspective. The data were collected by conducting an online survey of all engineering students in the College of Engineering, Informatics, and Applied Sciences at Northern Arizona University. The survey was scoped to identify the plethora of current experiences of students, explore related major duties and responsibilities, and self reported holistic competencies. This paper investigated to what extent these experiences shaped students’ professional identities, practice readiness, and motivated them to persevere through their studies. The findings of this work close-the-loop, and can be utilized to improve the activities of engineering career development offices across the world.

Introduction

Internships have long been recognized as valuable by students, employers, and career services professionals aiding job search, recruiting, and career development [1].

“For students and career services professionals, internships offer a range of benefits, including the opportunity to identify and clarify career direction, develop skills important to career readiness, and gain first-hand experience in the workplace. For employers, internships can serve as a valuable source of new hires, enabling the organization and potential hire to try each other out, thereby enhancing retention down the road as well. [1]”

Here we argue that it is not only internships, but also professional experiences including non-technical or those unrelated to one's major that provide value for students' career readiness. In this study, professional experiences were inclusive of working for an organization in a position related or unrelated to one's major (paid or volunteer); doing research in a position related or unrelated to one's major (paid or volunteer); and other personal experiences, like camps (summer or any other time of year); sports teams, practice, competitions; art and/or music lessons and practice; religious studies; or self-taught skills such as learning computer coding, flying drones, fixing a car, woodworking, etc.

Experiences guide who we are, both in terms of personal identity and in terms of preparation for the future workplace. In other words, we can think of readiness as a state of preparation [2], a competency, which includes both emotional and physical aspects [3]. Consequently, work readiness is a measure of one's ability to complete assignments in accordance with the provisions, without experiencing major setbacks, to provide maximum results [3]. In this paper, we will refer to this work readiness, as practice readiness by placing an emphasis on "practicing" engineering, from a holistic perspective.

Literature Review

Areas of research literature relevant to this study include the topics of: the need for practice ready graduates, discrepancies between industry expectations and students' practice readiness, and standard instruments to measure students' practice readiness.

The need for practice ready graduates

To ensure international competitiveness in today's knowledge-based economy, all stakeholders expect university graduates to be "practice ready" [4]. To this end, since the 1970s, ABET has emphasized the student Outcomes (Criterion 3), which are expected to prepare graduates to seamlessly enter into the professional practice of engineering. ABET student outcomes are organized into seven dimensions. Most professors agree that these are important for engineering graduates [5]. Four of these dimensions are directly assessed here: problem solving (1); problem solving in a global, environmental, and societal context (2), effective communication (3), ethical responsibilities (4), teamwork and leadership (5), and the ability to acquire and apply new knowledge (7).

In Hugo's 2010 review of "Educating Engineers - Designing for the Future of the Field by Shepherd et al. (2009)" it is argued that there is a disconnect between engineering education and practice [6]. There has been a slow transformation of engineering education in response to profound evolution within the engineering profession, which has proved insufficient to prepare engineering students to deal with the challenges that current engineering practice demands. They urged a focus on professional practice within undergraduate engineering education curricula and a move away from the traditional curriculum (i.e. linear, deductive, structured one-right answer problems) to open-ended scenarios and experiences [6].

Several studies have articulated employer expectations for engineering graduates; practice ready graduates are more likely to have higher potential job performance, success, promotion, and career advancement [7, 8].

To add to this, the unexpected impact of the pandemic, and drastic shift to work from home, requires these new graduates to be practice ready more than ever before.

The discrepancy between industry expectations and students' practice readiness

Engineering student placement experience is recognized as increasingly important in their pre-employment education [9]. Why? The transition from senior year at a university into the workforce is a critical time for graduates, especially those with limited previous work experience [10]. The workplace reality requires students to not only have strong academics, but also be equipped with skills and perspectives typically gained outside of academic settings [3].

The concerns about discrepancy between the industry expectations and graduate practice readiness has been alluded to for general engineering [6], studied within software engineering [11], civil engineering [10], and chemical engineering graduates [12].

Literature has shown that there are a number of approaches higher education institutions have taken to improve their graduates' practice readiness, some of which have enhanced students' employability through developing generic skills or holistic competencies [4]. Additionally, some institutions have implemented capstone design classes, in an effort to meet ABET criteria on student outcomes and address concerns regarding graduates ill-prepared for industrial practice [5, 13].

To enhance students' employability skills, career identity and practice readiness; Green, Carbone, and Rayner (2019) argued for more flexible and accessible alternatives to the traditional curriculum [14]. Carbone et al. (2020) identified a range of programs and practices, in addition to placements and internships, including practicums, projects, field trips, and site visits within the plethora of activities that can be used to develop practice ready graduates by expanding students' perceptions of their career and professional identity [13].

Another example of flexible and accessible alternatives is student organizations. Kurniawaran et al. (2020) argued the positive and significant impact of students' involvement in organizations on work readiness [3]. In addition to providing training to work within multidisciplinary teams, active organizations help students foster an independent attitude, confidence, discipline and responsibility, improve time management skills, improve communication skills, expand their networks, develop insight, increase awareness and sensitivity to the community and to the environment, and foster critical, productive, creative, and innovative abilities (Kurniawan and Puspitaningtyas (2013) in [3]), [15].

Standard Instruments of “Practice Readiness” and their Limitations

Little is known about how well prepared final-year engineering students perceive themselves to enter the workforce [10]. There is increasingly recognizable value in identifying an instrument to measure student practice readiness. The level of graduate readiness for practice has been studied for medicine [16], nursing [17, 18], finance and accounting [19]; and economics [3]. There are numerous sample internship exit interview surveys available online [20-22]. Although these instruments might give insight into participants’ practice readiness, they have not been validated; therefore their reliability is questionable.

Prikshat et al. (2019) stated that there is no uniform scale for accurately documenting graduate work-readiness within the context of escalating and changing needs in education and practice. They reviewed several existing graduate work-readiness and skill assessment instruments, and concluded that some lack to assess the personal attributes and personality traits, which may be associated with generic skills graduates are expected to possess [23]. They then measured work-readiness of the graduates, who are seeking their first full-time job in industry, with an integrated competencies model for Asia-Pacific specific region [23].

O’Brien et al. (2012) looked at final year civil engineering students in Melbourne, Australia by including 9 dimensions, engineering fundamentals: knowledge, analysis, and design; modern tool usage; engineering responsibility to society and environment; ethics; individual and teamwork; communication; project management and finance; as well as lifelong learning; and coping skills. The survey included 52 items, on a 6-point scale ranking their responses with respect to how well their degree had prepared them in relation to each specific topic: from “very inadequately” to “very adequately”. Their sample size consisted of 19 responses [10].

The study by Male et al. (2011) derived a statistical 11-factor generic engineering competency model by comparing experienced engineer’s perspectives to the industry leaders’ in Australia. The resulting model included communication, teamwork, self-management, professionalism, ingenuity, management and leadership, engineering business, practical engineering, entrepreneurship, professional responsibilities in addition to applying technical theory [24]. The teaching methods and learning opportunities that develop these generic competencies within an engineering framework are recommended for engineering students to prepare for practice [24].

The studies mentioned above were either country specific, not engineering, or discipline specific with limited sample size [3, 10, 16-19, 23, 24]. Some of these studies employed a comprehensive instrument [10, 23], for which survey fatigue is known to be a problem. The scales presented above have the potential to systematically measure practice readiness, but they suffer from some limitations. There is a need to further validate a generic “practice-readiness” scale, applicable to all disciplines through a lens of an inclusive definition of “professional experiences”.

Based on this previous literature, we have identified that future work investigating this connection between professional learning and student practice readiness should make use of a validated scale. Chan and Luk (2020) have developed and validated such a scale ($N > 2000$)

undergraduate students) and have agreed to provide their full survey for use in this study. We took this one step further, as we also explored underrepresented (gender and URM) students' perspectives and experiences.

Methodology

Students from multiple disciplines in engineering and applied sciences within the college were surveyed online and asked to identify their experiences. The participants were recruited using a college-wide all-student Listserv, and also delivered by the authors, in their respective classes. Total of 282 students started the survey, however 180 respondents completed its entirety. The survey asked students to self-identify their program of study, how many semesters left to graduate, their experiences, job duties, professional identity, resilience and finally, holistic competencies as identified by Chan et al. (2017) and validated by Chan and Luk (2020). Participants provided short answers for some questions, selected responses using a five-point likert scale that ranged from “Strongly Disagree” to “Strongly Agree”; “Not at All Confident” to “Very Confident”; or “Very Poor” to “Very Good.” All analyses were done within Matlab R2020b, leveraging built in functions for comparisons and regression.

Chan and Luk (2020) argued that questions pertaining to demographic information prior to the evaluation items may draw participants' attention to stereotypes and their awareness of minorities. In order to eliminate bias, we omitted demographic information questions from the survey, but did request the respondent's email. This was then used to match the student's survey responses to institutional records of gender and race/ethnicity [25]. The team was composed of two faculty members with the college career development professional. The career development professional managed all data collection, matching of demographics to responses, and data de-identification before sharing any of this data with the faculty members to perform the analysis.

Analysis revealed that out of the 180 complete responses, 63 (35%) were female and 117 (65%) were male. See Table 1 for demographics, including race/ethnicity and gender comparing our study cohort to the whole college; relative percentage of each is similar.

This study sought to answer the following research questions (RQ):

1. *To what extent do students feel they possess the holistic competencies to be practice ready?*
 - 1.a. *Does the self-reported level of practice readiness correlate with how many semesters are left before graduation?*
 - 1.b. *To what extent do professional experiences impact students' practice readiness?*
 - 1.c. *Does the self-reported level of the practice readiness differ across various professional experiences? Which experience is the most impactful?*
2. *To what extent do professional experiences influence students' professional identity?*
3. *How do RQ1 and RQ2 differ for underrepresented student groups (gender and URM)?*

Table 1: Distribution of race/ethnicity and gender within the study and within the college.

<i>Race / Ethnicity</i>	Study Cohort		College	
	<i>Number</i>	<i>Percentage</i>	<i>Number</i>	<i>Percentage</i>
White	107	59%	1246	54%
Hispanic / Latin x	36	20%	459	20%
Two or More	11	6%	139	6%
International	8	4%	241	10%
Asian	7	4%	57	2%
American Indian / Alaska Native	6	3%	80	3%
Black / African American	4	2%	49	2%
Not Specified	1	1%	40	2%
Hawaiian / Pacific Islander	0	0%	12	1%
<i>Gender</i>	<i>Number</i>	<i>Percentage</i>	<i>Number</i>	<i>Percentage</i>
Female	63	35%	487	21%
Male	117	65%	1836	79%

Data Analysis and Results

Table 1 shows the distribution of our study cohort as compared to the overall college both in terms of ethnicity and gender. The fractional representation of ethnicity and gender in our study cohort is similar to the overall college. Given this similarity, we can expect our analysis to be representative of the college, and findings to then be applicable to the college.

To answer RQ1: To what extent do students feel they possess the holistic competencies to be practice ready. We analyzed the mean and standard deviation for all competencies listed by Chan et al. (2017) as shown in Table 2. There are 27 competencies listed under question #14 of the survey. The full survey is included in the Appendix.

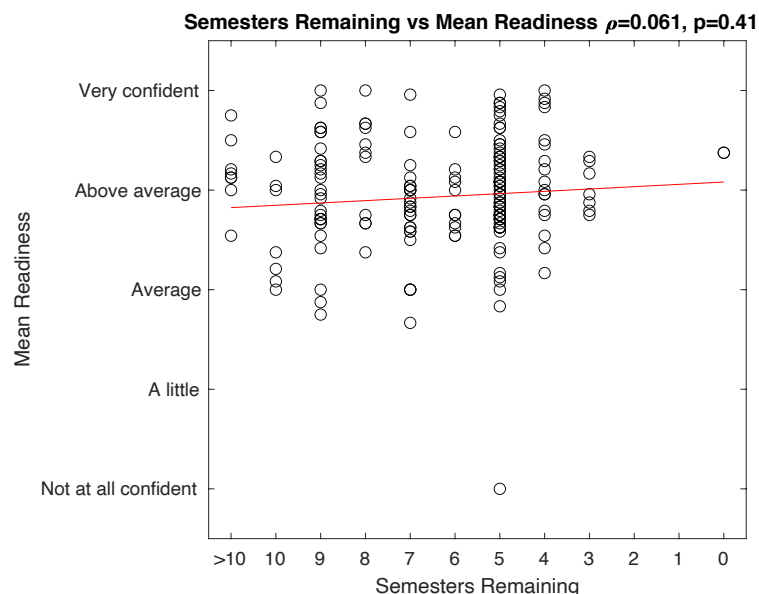


Figure 1: Shown here are the per student mean readiness (*y-axis*) vs the number of semesters remaining (*x-axis*), indicated by each circle. The red line shows a first order regression model to fit the data. This regression shows a positive trend, though data from students with two or three semesters remaining was limited.

Table 2: Mean and standard deviation of student respondent self evaluations on 5 point likert scale of confidence for each holistic competency question. Here, 1 is “Not at all confident,” 3 is “Average,” 5 is “Very confident.”

Mean Response	Standard Deviation	Question
3.406	1.102	Self-confidence
3.472	0.868	Persuade others
3.683	0.936	Have information technology skills
3.767	0.992	Build and maintain relationships
3.728	0.927	Motivate others
3.817	0.972	Lead others
3.844	0.991	Understand your own culture
3.856	0.992	Be able to develop clear goals, self-direction, and self-discipline
3.889	0.804	Express and receive ideas clearly
3.889	0.776	Have the ability to effectively find, evaluate, and use information
3.894	0.948	Be responsible for team members
3.978	0.871	Communicate with others effectively in different contexts
4.011	0.997	Have a sense of citizenship and responsibility to contribute to the society
4.011	0.891	Be aware of personal strengths and weaknesses
4.044	0.775	Analyze and evaluate an issue
4.072	0.812	Have numerical skills
4.111	0.915	Be aware of ethical responsibility
4.139	0.746	Identify problems
4.189	0.775	Think critically
4.278	0.819	Appreciation for others
4.278	0.777	Be reliable
4.294	0.789	Consideration for others
4.322	0.816	Understand and respect other cultures and perspectives
4.378	0.853	Respect individual differences and preferences

We calculated the mean response, and the standard deviation, then listed the mean for each holistic competency from lowest to highest (Table 2). The students report that they are most considerate of others, understand and respect other cultures and perspectives, and respect individual differences and preferences.

The next question we sought to answer was “RQ1a: Does the self-reported level of practice readiness correlate with how many semesters are left before graduation?” To do this, we regressed the overall mean readiness of an individual’s mean response to all questions, against the number of semesters remaining. We have reported the fit ($\rho = 0.061$). Figure 1 shows the mean readiness across the number of the semesters left. As shown on Figure 1, as the number of semesters remaining decreases (close to graduation), the overall readiness increases. This correlation is small, and not statistically significant, yet there appears to be a small positive gain towards graduation moving the line above average readiness. Additional connection was observed when RQ1b was explored. As Table 3 reflects, there appears to be a strong correlation ($\rho = 0.913$, $p = 0.087$) between the mean readiness response and the number of professional experiences a student has. A t-test was used to compare overall (mean) readiness of the students

who had no professional experiences to the group of students who had one or more experiences ($p = 0.236$, $t = 1.189$, $df = 280$).

RQ1c explored “Does the self-reported level of the practice readiness differ across various professional experiences? Which experience is the most impactful?” A total of 498 experiences were reported for this question on the survey from a total of 282 respondents. The analysis of RQ1b was repeated for the mean readiness according to type of experience. Table 4 shows 66 students reported having none of the professional experiences listed in the survey, 139 reported exactly one, 55 reported exactly two, 19 reported exactly three, 2 reported exactly four and 1 reported exactly five professional experiences, Table 5 shows the clear difference between students with no experience (3.786 ± 1.001) versus some experience.

Table 3: Mean and standard deviation of aggregate student response to self evaluations of holistic competency questions, grouped by the number of experiences.

Mean Response	Standard Deviation	N	Experience (grouped to number of experiences)
3.786	1.001	66	No Professional Experiences
3.920	0.736	216	One or More Professional Experiences
3.879	0.732	139	1 Professional Experience
3.923	0.796	55	2 Professional Experiences
4.273	0.467	19	3 Professional Experiences
4.000	1.414	2	4 Professional Experiences

} $p = 0.236$, $t = 1.189$, $df = 280$

Table 4: Mean and standard deviation of aggregate student response to self evaluations of holistic competency questions, grouped by the type of experiences.

Mean Response	Standard Deviation	N	Experience (grouped to number of experiences)
3.786	1.001	66	None of these experiences
4.016	0.745	115	Worked for an organization (paid or not) : <i>Related to major or not</i> *
4.100	0.712	66	Worked for an organization in a position related to my major
4.000	0.756	66	Worked for an organization in a position NOT related to my major
4.000	0.866	38	Did research (paid or not) : <i>Related to major or not</i> *
4.000	0.926	31	Did research in a position related to my major
3.667	0.577	10	Did research in a position NOT related to my major
3.911	0.709	146	Personal experiences, like Camp (any other time of year)

* : aggregate measures that combined the responses to both “related to my major” and “NOT related to my major.”

Table 5: Mean and standard deviation of aggregate student response to self evaluations of professional identity, grouped by the number of experiences. As can be seen here, there exists a statistically significant difference between the No Prof. Exp. and the One or More Prof. Exp. groups; $p = 0.004$.

Mean Response	Standard Deviation	N	Experience (grouped to number of experiences)
3.349	0.948	66	No Professional Experiences
3.768	1.063	216	One or More Professional Experiences
3.696	1.051	139	1 Professional Experience
4.000	1.000	55	2 Professional Experiences
3.636	1.286	19	3 Professional Experiences

} $p = 0.004$, $t = 2.870$, $df = 280$

To answer RQ2, “To what extent do professional experiences influence students' professional identity?”, both qualitative and quantitative data were analyzed. We qualitatively grouped responses into themes and categories and have reported the frequency distribution of these response types. We also applied quantitative methods, descriptive and t-test, to analyze student confidence (instead of preparedness).

Students provided responses to the prompt “Give a brief description of the experiences you have found most influential in your career choice(s).” Participants were able to respond to this question up to four times if they had or wanted to list more than one experience. The maximum response count was 171 for the first experience listed, which can be argued as possibly the most impactful experience on students' professional identity.

The responses for the prompt were analyzed using thematic analysis [26]. A single coder initially conducted the analysis of the responses, starting with a set of a priori codes that came from the development team’s vision of the survey. Codes were categorized by student responses with a focus on diverse professional experiences listed. As the analysis progressed, initial codes were modified, condensed and new codes emerged to better capture students’ professional experiences. A second coder then confirmed codes, and later the team came to a consensus on final codes, themes and categories (Table 6). The frequency distribution of the categories is shown in Figure 2. According to the student responses, the most influential experiences students have, are in the category of preparation. To validate the accuracy of research findings, the researcher employed two strategies recommended by Creswell [27]: peer debriefing (discussions with the team during data collection and analysis) and consideration of discrepant information that ran counter to the themes. The data was continually reexamined during analysis as patterns and themes emerged [28].

For RQ2, the quantitative analysis depends on “How confident are you that you are on the right track for your desired future career?” question in the survey. Responses were coded according to a 5 point Likert scale (1: not at all, 2: a little, 3: average, 4: above average, 5: to a great extent). The data is analyzed across the number of the semesters before graduation, and the type of professional experience reported. Table 5 shows the Mean Self-assessed Professional Identity.

Figure 2: Shown here are the responses to the question “ Give a brief description of the experiences you have found most influential in your career choice(s).” The emerged categories, and their relative occurrences, are reflected in this Venn diagram.

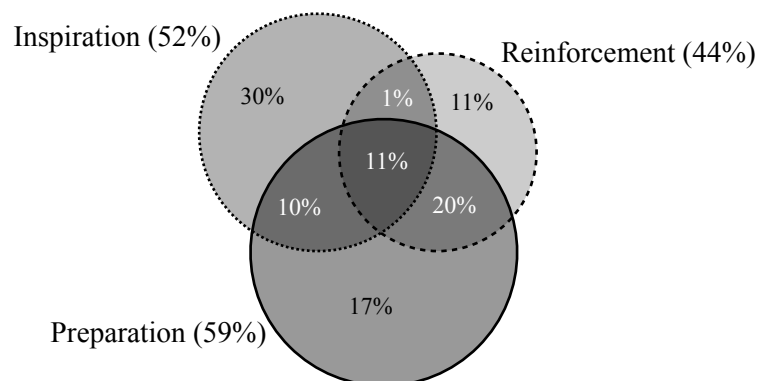


Table 6: Codes and themes emerging from participants' experiences they found most influential in their career choices. One can see the set of codes is quite diverse.

Category	Codes and Subcodes	Themes
Reinforcement	Personal Counseling Luck and Opportunity Teachers Parents, Family members	Encouragement
Reinforcement & Preparation	Class and Teacher Engineering/Astronomy Summer Camp Competition-Robotics	Encouragement Supported with Preparation
Preparation	Teaching Assistant for a Class Shadowing Repair Jobs Hands on Work	Major Related Work
	Navy, NREIP NJROTC	Military Experience
	Research	Research
	Athletics Boy Scouts	Extracurricular Activity
Preparation & Inspiration	Fixing Cars Restoration on Cars Worked on Engines Sustainability and Conservation Service Trip Competitive Motorsports	Working in a Supported Environment Towards a Purpose
Inspiration	Working for a Cause Working for an Non-Governmental Organization	Non-Major Related Work
	Volunteer Work Missionary Work Service Trip	Service
	Hobby Building Something, Legos Video Games Riding Roller Coasters Puzzles Travel Visiting Institutional Centers (e.g. Congress) International Travel Experience Love of the Outdoors Camps	Personal Interest
	Negative Experience Disability	Significant Negative Event/Experience
Inspiration & Reinforcement	Observation of the Real World Work Taking an Interest Survey	Introspection
Reinforcement, Preparation, & Inspiration	Hands on Engineering School/Camp Related Activity Internship at an Engineering Firm Repairs with Family Support Being Part of Society of Women Engineers	Scaffolded

In RQ3, we further sought to investigate the findings for RQ1 and RQ2 by cross analyzing students' underrepresentation (URM) status and gender descriptors. We described the incidence of experiences as dichotomized by gender and URM status. We then performed a t-test for each paired group, paired by gender and URM status. Results of the gender analysis are reflected in Table 7; results of the URM analysis are not included because no statistically significant difference was found.

Table 7: Shown in this table are the mean, and standard deviations of readiness for each group, male and female, in sum and separated into No Prof. Exp. and Some Prof. Exp. groups. Between each set of means and standard deviations, are the t-test results of those comparisons. All paired comparisons were found not to be significant, except the paired comparison of males with some professional experience and females with some professional experience. This demonstrates that the impact of those experiences on female students is significant.

Male		Female	
All (117)	3.927 ± 0.590	$p = 0.128,$ $t = 1.527,$ $df = 178$	4.058 ± 0.450 All (63)
No Prof. Exp. (26)	3.946 ± 0.799	$p = 0.798,$ $t = 0.257,$ $df = 40$	3.888 ± 0.509 No Prof. Exp. (16)
	$p = 0.860,$ $t = 0.177,$ $df = 115$		$p = 0.081,$ $t = 1.775,$ $df = 61$
Some Prof. Exp. (91, 77.8%)	3.922 ± 0.521	$p = 0.030,$ $t = 2.199,$ $df = 136$	4.115 ± 0.418 Some Prof. Exp. (47, 74.6%)

† 3 students provided no email, no gender could be identified for them.

Discussion

Our results show the interrelatedness of student practice readiness, holistic competencies, professional experience type, the number of experiences, and number of semesters before graduation (RQ1).

The mean scores of self-reported individual holistic competencies ranged from 3.406 ± 1.102 (Self Confidence) to 4.378 ± 0.853 (Respect individual differences and preferences). The top three competencies included the following: (1) Respect individual differences and preferences (4.378 ± 0.853); (2) Understand and respect other cultures and perspectives (4.322 ± 0.816); (3) Consideration for others (4.294 ± 0.789); see Table 2.

As one would expect, the overall readiness of the participants appeared to increase as the number of the semesters left to graduate decreased. However, this finding could be limited by the small sample size, especially given the absence of representation from students with one or two semesters remaining.

Students having no professional experience reported the lowest mean readiness (3.786 ± 1.001). Professional experiences do prepare students for the job (practice readiness), and increase their self confidence (reported by the mean readiness). The impact is even greater when students move from having two professional experiences (3.923 ± 0.7896) to three professional experiences (4.273 ± 0.467), they become closer to being “very confident”; see Table 3.

The mean readiness scores reported by students with different experiences showed small differences amongst professional experiences. Among the students who had professional experiences, the highest mean readiness was reported by the students, who worked for an organization (paid or volunteer) in a position related to their major. Doing research (paid or volunteer) in a position not related to one’s major seems to be the least helpful in building practice readiness. This analysis also revealed that among the respondents to the survey, students doing research (paid or volunteer) is the minority group ($N = 10$ vs $N = 31$ to 146); see Table 4.

The next question (RQ2) explores how one's professional identity connects to the professional experiences they have had. The single most common response was fixing or working on cars. Open ended responses indicated the distribution of professional experiences across categories; we see Preparation experiences were reported as most overall impactful (59% total) on one’s professional identity, followed closely by Inspiration (52%), and finally Reinforcement (44%) experiences. However, Inspiration was listed as the most common single factor of these experiences, which impacted students’ professional identities (30%). See Figure 2. Further, as shown in Table 6, there is a very eclectic set of experiences that leads a person to an engineering career.

In our study cohort, having two professional experiences produced the highest mean confidence (4.00 ± 1.0 , $N = 55$). It is valuable to note though that the sample size for having three professional experiences is relatively small, consisting of only 10 respondents, which may be impacting the drop on the mean. Although there is a medium correlation between the number of professional experiences and the professional identity, it is not statistically significant ($\rho = 0.564$, $p = 0.436$).

When data was examined across for students with no experience (3.349 ± 0.948 , $N = 66$) versus students with some experience (3.768 ± 1.063 , $N = 216$), revealed $p = 0.004$, $t = 2.870$, $df = 280$. Having “a professional experience” whether it is paid, or volunteer, research or internship, related to major or not, has a statistically significant impact on one’s professional identity as compared to students with no such experiences.

Using the lenses of URM status and of gender, our findings were in alignment with what we had expected, but the magnitude of these effects was smaller than anticipated. When we considered who has had one or more of these professional experiences, we found that 72% of URM students and 79% of non-URM (white) students had an experience(s). While 8 in 10 of the non-URM students have had such an experience, only 7 in 10, or 87.5% ($\frac{7}{8}$) as many non-URM students, are getting gaining experiences. We found no statistically significant differences between URM and non-URM groups, nor between each group divided into those with and without experiences.

When considering gender, there exists no statistical difference between the readiness of males or females who have had no professional experience. However, there does exist a statistically significant difference between the two genders for those who have had some number of professional experiences. This demonstrates that the impact of those experiences on female students is noteworthy.

Conclusion

This study considered survey responses from 282 students in the college of engineering, of which 180 provided complete responses. We looked at student practice readiness by using a validated perceived holistic measures instrument, developed by Chan et al. 2017 and Chan and Luk 2020, which is aligned with National Association of Colleges and Employers (NACE) and current literature. The demographics representation of the respondents closely matched that of the college. Respondents collectively self ranked their confidence for each holistic competency between average to very confident. Within our inclusive definition of a professional experience, we have seen that having any of these experiences makes a difference as compared to not having any such experiences. There is suggestive evidence of a strong, and positive correlation between a student's readiness and the number of professional experiences they have. Further there exists a positive, and significant, correlation between the number of professional experiences and a student's professional identity. Students with any professional experiences have a significantly higher self rated professional identity than those without any such experiences. For our students to identify themselves as engineers, having two experiences proved most constructive.

The URM students in this study report a very similar readiness to non-URM students, in contrast with NACE. This suggests that the many efforts to engage underrepresented minority students at this university have been fruitful and effective. However, there still exists an opportunity for improvement within the college, as underrepresented minority students are 87% as likely as non-minority students to have professional experiences before graduation. Prior to an experience, there is no statistically significant difference between female and male students for practice readiness. Following an experience, there is a statistically significant difference between the genders. This supports the hypothesis that experience matters, specifically for females.

This work shows that, inspiration is almost as impactful as preparation or reinforcement experiences on students' career choices. Therefore, we suggest that career development efforts can help remedy the discrepancy between industry expectations and students' practice readiness by emphasizing all kinds of experiences matter, related to one's major or not. Further, we suggest targeted recruitment of females into eclectic experiences to close the career readiness gap between the genders.

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Appendix 1: The survey used in this study was a combined version of Chan and Luk (2020) with the addition of context questions to address students' experiences. Questions (Q#) are noted with (r) as required, (o) as optional, † as dependent on "yes" to Q4 *a-d*, and ‡ as dependent on "yes" to any of Q4. Multiple choice questions are marked with a ✓. Subquestions are noted with a *lightened italic number or letter*. * Q10 scale applies to all subquestions.

Q1 (r)	School email: <i>[used for campus self reported gender and ethnicity]</i>
Q2 (r)	Please indicate your program of study <i>[menu of programs specific to institution]:</i>
Q3 (r)	How many semesters (after this one) do you anticipate there are before your graduation?
	0
	1
✓	...
	10
	Likely more than 10
Q4 (o)	Which of the following experiences can you identify having had a significant impact on you (outside of classwork), to the extent that you now feel competent to teach it to someone else?
a	Worked for an organization (paid or volunteer) in a position related to my major
b	Worked for an organization (paid or volunteer) in a position NOT related to my major
c	Did research (paid or volunteer) in a position related to my major
d	Did research (paid or volunteer) in a position NOT related to my major
e	Personal experiences, like Camp (summer or any other time of year), sports teams/practice/competitions, art/music lessons/practice/group, religious studies or group, self-taught skills (learning computer language/coding, flying drones, fixing a car, woodworking...)
f	Other <i>[open ended response]</i>
Q5 (†)	Please provide the name of the organization(s) you completed your experience(s) with:
	<i>[open ended response]</i>
Q6 (†)	Please copy/paste your job description or summarize your duties here:
	<i>[open ended response]</i>
Q7 (‡)	Give a brief description of the experiences you have found most influential in your career choice(s). Ex. I rebuilt a car engine during high school / I enjoy flying drones in my spare time / I participated in a summer coding camp/ I've been training and riding horses since I was 6 / my family and I go hunting every year / I have been playing the saxophone since I was 8 / my high school science teacher said I have the skills to follow this career path/ I went on a year-long mission for my church / I have participated in 3 mock United Nation competitions...
1	First Experience <i>[open ended response]</i>
2	Second Experience <i>[open ended response]</i>
3	Third Experience <i>[open ended response]</i>
4	Fourth Experience <i>[open ended response]</i>
Q8 (r)	Please indicate your agreement with the following statement: I intend to complete an engineering degree within the next five years.
✓	Strongly Disagree Disagree Neutral Agree Strongly Agree
Q9 (r)	How confident are you that you are on the right track for your desired future career?
✓	Not at All Confident A Little Average Above Average Very Confident

Appendix 1: The survey used in this study was a combined version of Chan and Luk (2020) with the addition of context questions to address students' experiences. Questions (Q#) are noted with (r) as required, (o) as optional, † as dependent on "yes" to Q4 *a-d*, and ‡ as dependent on "yes" to any of Q4. Multiple choice questions are marked with a ✓. Subquestions are noted with a *lightened italic number or letter*. * Q10 scale applies to all subquestions.

Q10 (r) How would you assess your current level in the following competencies? *		✓	<i>Very Poor</i>	<i>Poor</i>	<i>Average</i>	<i>Good</i>	<i>Very Good</i>
<i>a</i>	Express and receive ideas clearly						
<i>b</i>	Communicate with others effectively in different contexts						
<i>c</i>	Build and maintain relationships						
<i>d</i>	Persuade others						
<i>e</i>	Motivate others						
<i>f</i>	Lead others						
<i>g</i>	Be responsible for team members						
<i>h</i>	Be reliable						
<i>i</i>	Identify problems						
<i>j</i>	Think critically						
<i>k</i>	Analyze and evaluate an issue						
<i>l</i>	Have the ability to effectively find, evaluate, and use information						
<i>m</i>	Have information technology skills						
<i>n</i>	Have numerical skills						
<i>o</i>	Be aware of ethical responsibility						
<i>p</i>	Respect individual differences and preferences						
<i>q</i>	Have a sense of citizenship and responsibility to contribute to the society						
<i>r</i>	Understand your own culture						
<i>s</i>	Understand and respect other cultures and perspectives						
<i>t</i>	Be aware of personal strengths and weaknesses						
<i>u</i>	Be able to develop clear goals, self-direction, and self-discipline						
<i>v</i>	Self-confidence						
<i>w</i>	Consideration for others						
<i>x</i>	Appreciation for others						

Impact of Faculty Behaviors on Student-Faculty Rapport: A Multi-Institutional Study

Dr. Fethiye Ozis P.E., Carnegie Mellon University

Dr. Fethiye Ozis is an Associate Teaching Professor in the civil and environmental engineering department at Carnegie Mellon University. Dr. Ozis holds a B.S. in environmental engineering from the Middle East Technical University, Ankara, Turkey and M.S. and Ph.D. degrees from the University of Southern California. Dr. Ozis is a licensed Professional Engineer, Environmental, in Arizona. She is also an ENVISION Sustainability Professional. Before joining CMU, Dr. Ozis was a faculty member at Northern Arizona University.

Dr. Kyle Nathan Winfree, Northern Arizona University

Dr. Winfree is the Associate Director for Undergraduate Programs in the School of Informatics, Computing, and Cyber Systems at Northern Arizona University. His research focuses on wearable technologies as applied to health assessment and rehabilitation.

Dr. Corinna Marie Fleischmann P.E., United States Coast Guard Academy

CAPT Corinna Fleischmann is a licensed Professional Engineer with military, academic and research experience in water resources engineering, environmental engineering, coastal resiliency, construction project management and engineering education. CAPT Fleischmann is a career educator who has been a member of the US Coast Guard Academy (CGA) faculty since 2004. CAPT Fleischmann's passion for teaching and learning began early in her career. In 2007, she was certified as a Sustainable Building Advisor through the National Sustainable Building Advisors Program. In 2009, she began her PhD work at the University of Connecticut in Civil Engineering. Since earning her PhD in 2014 and has continued to seek out research opportunities collaborating with the United States Environmental Protection Agency (EPA) Office of Research and Development and Homeland Security on the Analysis for Coastal Operational Resiliency (AnCOR) project to assess the USCG's preparedness to respond to a wide-area biological contamination incident impacting stormwater infrastructure and with CGA and USCG Health, Safety and Work-Life (HSWL) to develop COVID-19 sewage surveillance protocols. CAPT Fleischmann is actively involved with the American Society of Civil Engineers (ASCE) Excellence in Civil Engineering Education (ExCEED) program and spends time each summer mentoring civil engineering faculty at colleges and universities throughout the US. As a faculty member at CGA, she expanded and modernized the Civil and Environmental Engineering curriculum with the development of elective courses in Water Resources Engineering and Coastal Resiliency. She served as the Department Head of Civil and Environmental Engineering 2017-2021 and is currently serving in a new role as the Dean of the School of Leadership and Management.

Dr. Scott R Hamilton P.E., York College of Pennsylvania

Scott Hamilton is a Professor of Civil Engineering at York College of Pennsylvania. He is a registered Professional Engineer and has both a MS and PhD in civil engineering and a Masters in engineering management from Stanford University and a BS from the United States Military Academy, West Point. He is a retired US Army Corps of Engineers officer who has had assignments in the US, Germany, Korea, and Afghanistan. During his military career he spent over 10 years on the faculty at the US Military Academy at West Point teaching civil engineering. He also served as the Director, Graduate Professional Development at Northeastern University's College of Engineering. He is the recipient of the 2021 NSPE Engineering Education Excellence Award and the 2019 ASCE Thomas A Lenox ExCEED Leadership Award.

Dr. Kaitlyn T Hanley P.E., New York University Tandon School of Engineering

Dr. Hanley is an Associate Professor of Environmental Engineering at New York University, teaching courses in environmental chemistry and microbiology, fundamental environmental engineering laboratory



techniques, environmental engineering process design, and the senior environmental engineering capstone design experience. She is a licensed Civil Engineer (CA) with over 17 years of specialized academic and industry experience, specializing in stormwater management, watershed-based planning, and microbial water quality engineering. In addition to her teaching and mentoring responsibilities, Dr. Hanley is working to expand the undergraduate Environmental Engineering program at NYU Tandon.

Dr. Tanya Kunberger P.E., University of Pittsburgh at Johnstown

Dr. Kunberger is Division Chair for Engineering and Computer Science at the University of Pittsburgh Johnstown.

Dr. Monica Palomo P.E., California State Polytechnic University, Pomona

Professor

B.S. Civil Engineering, University of Guanajuato, Gto, Mexico, December 1999, summa cum laude.

M.S. Civil Engineering, Kansas State University, Manhattan, KS, May 2003.

PhD. Civil Engineering, Kansas State University, Manhattan, KS, May 2008.

Dr. Palomo is currently a Professor in the Civil Engineering Department at California State Polytechnic University, Pomona (Cal Poly Pomona). In this position, Dr. Palomo is responsible for teaching courses such as Introduction to Civil Engineering; Hydraulics; Water and Wastewater Treatment; Groundwater Mechanics; Research Experience of Undergraduate Students; and Engineering Outreach Service Learning courses, among others. She is also a faculty advisor for the California Water Environment Association (CWEA), and Engineers Without Borders (EWB) student chapters. Additionally, Dr. Palomo is the CE Water Analysis laboratory director and coordinates all teaching, research and safety training activities in the engineering laboratory. Dr. Palomo conducts research in surface water quality improvement via natural treatment systems, water and wastewater treatment processes, and water education. She is involved in outreach programs for K-12 students to increase the participation of Hispanic female students in STEM fields

Dr. Jennifer Retherford P.E., The University of Tennessee, Knoxville

Dr. Retherford is an alumna of the University of Nebraska, Omaha, and received her graduate degrees from Vanderbilt University. She currently teaches a variety of courses supporting the department of Civil & Environmental Engineering at the University of

Dr. David A Saftner, University of Minnesota Duluth

David Saftner is an Associate Professor at the University of Minnesota Duluth. He received a BS in Civil Engineering from the United States Military Academy and MS and Ph.D. in Civil Engineering from the University of Michigan.

Dr. Camilla M. Saviz P.E., University of the Pacific

Camilla Saviz is Professor and Chair of Civil Engineering at the University of the Pacific. She holds B.S. and M.S. degrees in Mechanical Engineering from Clarkson University, an M.B.A. from the New York Institute of Technology, and a Ph.D. in Civil and Environmental Engineering from the University of California, Davis. She is a registered engineer in California.

Impact of Faculty Behaviors on Student-Faculty Rapport: A Multi-Institutional Study

Abstract

This research paper describes the findings from a multi-institutional exploratory study investigating student perceptions of positive rapport-building faculty behaviors. Numerous studies have identified that professors who can establish strong and positive rapport with their students have an immediate and beneficial impact on students' learning, attendance, engagement, motivation, and academic success, resulting in a positive long-term influence on retention. Building on previous work by Ozis and Winfree that ranked fifteen specific faculty behaviors fostering positive rapport at a single institution in the southwestern United States, this study includes perspectives from undergraduate and graduate engineering students across multiple institutions with different academic populations.

The objectives of this study are twofold: first, to identify the faculty behaviors that have a higher potential for positive influence on rapport, and second, to determine whether, or the extent to which, demographics (e.g., gender, age, GPA) affect students' perception of what constitutes positive student-faculty rapport.

This study reports on students' opinions across a diverse group of universities through a survey and recommends strategies for developing positive faculty-student rapport. Responses indicated that students value positive interpersonal behaviors such as respectfulness, understanding, kindness, and caring; professional behavior such as answering emails in a timely fashion; and beneficial pedagogical approaches such as conveying enthusiasm for the subject and providing relevant examples in class. The results of this work can be used broadly by faculty to enhance student learning and engagement through deliberate actions that develop positive rapport with students.

Introduction

Interactions between students and teachers in a classroom have the potential to positively or negatively impact the classroom environment [1]. Understanding the dynamics of student-faculty rapport is essential for creating a thriving learning environment. Ozis and Winfree [2] identified key faculty behaviors that foster positive rapport at a single southwestern state institution with a diverse group of college students including engineering and informatics. The professors' respect for students was consistently identified as a top priority across all analyzed respondent demographic categories, with first-year students valuing being known by name and female students appreciating office hours that were posted and consistently held.

This paper expands upon the work of Ozis and Winfree [2] in two distinct ways. First, this study includes multiple public and private institutions across the United States located in various geographical locations, each with unique academic populations. Second, this study surveyed students from different undergraduate academic years, and graduate students, as well as an assortment of engineering fields of study, including civil, environmental, transportation, chemical, biomolecular, and construction engineering and management.

Building on previous work, the objectives of this study were to:

- Identify faculty behaviors that have a higher potential for positive influence on rapport; and
- Determine whether, or the extent to which, demographics (e.g., region, type of institution, student's background, gender, status) affect students' perception of what constitutes positive student-faculty rapport.

This study has identified faculty behaviors and factors that help develop positive rapport in the classroom. The intent of using a multi-institutional approach is to develop recommendations that are current and applicable to a wide range of educational settings, ultimately contributing to improved learning outcomes and student retention across different educational settings.

Importance of Student-Faculty Rapport

Positive rapport between professors and students has been reported to improve student engagement, attendance, motivation, retention, and success within higher education settings [1] [3] [4]. In a seminal study, Lowman [5] presents interpersonal rapport as one of two dimensions required for teaching excellence, the other dimension being intellectual excitement. Some studies have related “rapport,” the agreed mutual respect and empathy between a professor and learner, to student persistence in academic performance. Numerous studies, including some summarized by Sybing [6], have evaluated the varied types of interactions between faculty and students, including a focus on specific encounters such as dialogue during active or passive class sessions, demeanor evident during lecture-style classes, and attitudes during office hours. Further, the research questions presented in the literature posit complex psychological relationships and emerging themes that identify rapport as a valuable element towards sociocultural experiences necessary to support diverse learners in higher education settings. Sybing [6] offered an additional key theme, namely, identifying the essential role of the connection between a faculty and student as a primary mechanism for knowledge transfer. Sybing [6] concisely summarized the work by Chickering and Gamson [7] that named the student-faculty relationship as a collaborative process necessary for successful learning. The concept of naming the relationship as a process emphasizes the evolution that will take place as the learner matures and gains knowledge while the educator adapts towards refining and affirming the knowledge transfer.

While many studies highlight the beneficial effects of positive rapport, some literature emphasizes the detrimental impacts of negative rapport. The highly cited work of Seymour and Hewitt [8] documented the “chilly” environment in engineering classrooms, much of which was related to student-faculty interaction. Several studies through the 1990s and early 2000s evaluated the retention issues in engineering as related to interactions and experiences in the classroom [9] [10]. Most students reported the learning environment as a significant reason for leaving engineering disciplines [11]. Findings from Vogt's [12] research linked poor faculty interactions to low retention rates among engineering students. Additionally, the study concluded that negative experiences with faculty significantly impacted students' self-efficacy and confidence. The results of negative rapport are consistently shown to be severe and should be considered detrimental to classroom engagement and management and to student learning and retention.

The mutually beneficial relationship built by positive rapport in a classroom requires action from both the learner and the faculty. The sense of rapport can decrease over time if the instructor does not continue to engage in authentic behaviors that elicit a sense of rapport from their students [13]. Felder and Brent [14] described differences among learners that could affect the

potential for building rapport, namely, students' learning styles preferences (characteristic ways of taking in and processing information), approaches to learning (surface, deep, and strategic), and intellectual development levels (attitudes about the nature of knowledge and how it should be acquired and evaluated). As summarized extensively by Wankat and Oreovicz [15], the work of Aubrecht [16] shows student ratings directly capture student opinions on rapport. Concepts such as accessibility [13] [17], availability [13], and approachability [16], [18] [19] have been cited as characteristics important to students, particularly while completing end-of-semester course evaluations. Faculty behavior and approach to teaching also determine rapport [13] [20]. Wankat and Oreovicz [15] initialize the effort for building rapport with the motto "know my name" [21] and cite additional work performed by Daly et al. [22] that prioritizes the straightforward task of knowing student names. The importance of positive interpersonal rapport in promoting teaching excellence is emphasized in the ASCE ExCEED Teaching Workshop and is included as a component of the ASCE ExCEED Model [23].

Although the literature extensively discusses the importance of rapport, outlines key behaviors, and identifies factors that matter, it fails to determine the most effective strategies for building positive rapport that account for student-specific factors, such as academic, logistical, and demographic variables. While most studies promote the value of rapport, they do not explicitly report detailed methodologies for how to effectively develop positive rapport [15] [24]. When methods are provided, they are often too general and lack specific guidance. Examples such as "complete a student information inventory form to collect a few basic traits about the student (hometown, major, etc.)" are excellent suggestions and are dispersed throughout the literature [25], but often do not clarify the types of details to solicit from students that are most effective for rapport building.

Building positive rapport creates personal connections and trust that results in a supportive classroom environment where students are willing to take intellectual risks and pursue their interests without fear of judgment. When students feel connected to their professors and peers, they recognize the value of the learning experiences, becoming more invested in their education to improve their lives and achieve their personal and professional goals. Professors who build rapport can tailor relevant and meaningful lessons to empower students to take an active role in their learning, reinforcing the impact and importance of their contributions. Identifying rapport-building strategies is crucial for both students and professors as they navigate the cross-cultural context of teaching and learning in college classrooms. In this context, faculty, who are content experts, could be supported by investing their energy into effective ways to establish and build the collaborative process of mutual, positive rapport. This paper provides faculty with strategies to develop positive relationships that can enhance student engagement and success.

Methods

The goals of this study were to identify faculty behaviors that have a positive influence on rapport and to determine whether, or the extent to which, demographics affect students' perceptions of what constitutes positive rapport. To address these goals, the authors developed a survey asking undergraduate and graduate students from nine different higher education institutions to provide feedback on faculty behaviors that constitute positive rapport.

In the first part of the survey, students were provided an open-ended question requesting three descriptors of a professor who left a positive influence or impression on them. The open-ended question was followed by a section comprising 21 faculty behaviors that were evaluated on a 5-

point Likert scale ranging from “Strongly Disagree” to “Strongly Agree.” Survey choices included the fifteen behaviors identified by Ozis and Winfree [2], together with six additional behaviors that reflected positive mentoring qualities, going beyond the fundamental behaviors typically associated with building positive rapport. The second part of the survey included questions related to the institution, year, gender identity, age, GPA, and other characteristics (e.g., being a first-generation college student, commuter, student-athlete, part-time student, or underrepresented group). Survey components are shown in Appendix A, were administered through Qualtrics, and distributed with the York University of Pennsylvania IRB approval (IRB# 24FA016).

The voluntary survey was administered by the study authors and faculty at their institutions. In most cases, the authors distributed the survey to students in classes they were teaching in the Fall 2024 semester. Several faculty awarded minimal extra credit for completing the survey, others posted the link as an announcement or assignment with no extra credit, and three faculty distributed the survey to all students in the program. Three hundred forty-two responses were received. Although it is possible that some respondents may have completed the survey multiple times, the authors thought that would be unlikely given the timing, with the survey being administered in the last two weeks of the semester. Survey responses were gathered and analyzed as described below.

Descriptor analysis

The survey asked respondents to “Think of a professor(s) who has or had a positive influence in your life or education. Provide (3) words or phrases that describe this person's interaction with you, or your observations of their interaction with others, that left an impression on you.” 967 total response items were received including 44 individual faculty names which were removed, yielding 923 usable responses. These responses were sorted and classified as follows:

- Descriptors were alphabetized.
- To simplify the analysis, some descriptors were reworded. For example, “Inspiring” and “Inspirational” were both classified as “Inspirational”.
 - 5% (44 descriptors) provided a unique word or phrase that seemed to be an appropriate fit with one of the 92 descriptors already in use. In these cases, the response was reclassified using one of the other descriptors, while maintaining the original intent. For example, “it’s ok to advocate for yourself” was replaced with “encouraging” and “cool and talkative” was replaced with “approachable”.
 - Some unique descriptors (e.g., “Gruff”) had no equivalent descriptor, so these were kept as given.
- A few respondents put multiple responses in each response box, yielding more than 3 descriptors, but only the first descriptor in each box was retained for the purposes of this study (e.g., “supportive and helpful” was classified as “supportive”).
- The descriptors were organized into four categories, classified as follows:
 - Personality traits (typically inherent to the personality, e.g., honest)
 - Supportive traits (external way they treat others)
 - Motivational traits (ways of motivating students)
 - Professional traits (professional/learned traits)
- Finally, the number of instances of each descriptor was counted. Results are described later in this paper.

The four categories listed above agreed closely with categories identified by Lowman [5], namely, Intellectual Excitement (IE), Interpersonal Rapport (IR), Commitment to Teaching (CT), or a General Positive Descriptor (PD), as follows:

- Personality traits -- General Positive Descriptors, Commitment to Teaching
- Supportive traits -- Interpersonal Rapport
- Motivational traits -- Intellectual Excitement
- Professional traits -- Intellectual Excitement

Although the survey specifically asked for behaviors that contributed to positive rapport, it is also likely that a professor who was identified as “Engaging” or “Inspiring” (Motivational traits) or “Effective” or “Knowledgeable” (Professional traits) could have a positive influence on a student’s life or education.

Multiple choice and response demographic analysis

The survey had a section to collect demographic information. This included the student's institution, year of study, age (by bracket), GPA (by bracket), and gender. Students were also asked to identify other unique characteristics such as being a commuter, a first-generation student, a part-time student, a student athlete, or if they identify as an underrepresented individual. The authors used these responses as the controlling categorical variables.

Student responses to the faculty behavior Likert-scale questions (prefaced by “It’s important to me that my professors...”) were analyzed as a whole with no categorical reduction, as well as based on six categorical variables: institution, year of study, gender identity, age, GPA, and personal demographics. This approach enabled the grouping of students into subsamples defined by each of these variables for more detailed analysis. For the purpose of analysis, Likert scale responses are assumed not just to be ordinal data, but also to be interval in nature. For each set of responses in these categories, we calculated a mean importance score from a weighted average using the following equation:

$$\text{Mean importance score} = \frac{(N_{SA} \times 5.0) + (N_A \times 4.0) + (N_{NAD} \times 3.0) + (N_D \times 2.0) + (N_{SD} \times 1.0)}{N_{SA} + N_A + N_{NAD} + N_D + N_{SD}}$$

Here, the number of participants who responded with a “strongly agree” (SA) to the faculty behavior Likert-scale question was multiplied by five, the number of participants who responded with a “somewhat agree” (A) was multiplied by four, the number of “neither agree nor disagree” (NAD) responses was multiplied by three, the number of “somewhat disagree” (D) responses by two, and finally the number of “strongly disagree” (SD) responses was multiplied by one. These products were then summed and divided by the total number of responses to that question. This process yields a score that reflects a weighted average.

Overall ranking, without regard for the categorical responses, was determined through the process of bootstrapping. For each question, the integer third of all responses were randomly selected (with replacement) and the mean importance score was calculated. This random sample and mean calculation was repeated 1000 times. The mean and the standard deviation for each question set of assessed mean importance scores were found. This process is a common approach to estimating metrics of the population from the sample at hand [26].

For each of the categorical variables (institution, year of study, gender identity, age, GPA, personal demographics), we binned the responses by the categorical variable of interest. For example, responses were sorted by the seven institutions. Likert-scale responses were counted and a mean importance score was calculated for each institution. Next, the mean and standard deviation across the institutions was calculated, where that mean reflects the score unbiased by the number of responses from each institution, given that each institution has equal weight in this average.

Once the equally weighted average for each faculty behavior Likert scale question was calculated, we identified the per question noise across all faculty behaviors with a mean and standard deviation of the standard deviations within each categorical variable. A threshold of the mean plus one standard deviation was used to identify the faculty behaviors where there was statistically significant disagreement of the importance of that faculty behavior. Under the assumption that the “most” important behaviors will receive the highest importance score, we selected the top three behaviors for further inspection. Further, we used the mean plus one standard deviation of the standard deviations across the categories for each question and categorical variable to identify the behaviors where there was the lowest agreement (highest variance in categorical mean) and offer a reflection on those as well.

Results

The survey considered in this analysis generated 342 responses. Tables 1 through 7 describe the number of responses from each of the seven institutions represented in this study, the respondents’ year of study, gender, age, GPA, and other factors. The responses were generally distributed among the types of institutions. Undergraduate students (first through fifth year) made up the majority of respondents (96%) with the remaining 4% identifying as graduate students. The percent of the survey population identifying as a woman (40.8%) exceeded the national average (33.4%) in civil engineering programs [27]. Fifty-eight percent of the student respondents list their age as 21 or older. Eighty-two percent of the respondents self-reported a GPA of 3.0 or higher. Finally, nearly one third of the students reported being a commuter, as defined by not living on or near campus, 22% being first generation college students, 2% being part time students, 16% being from an underrepresented group, and 15% being student athletes. The remaining 35% did not identify with any of those characteristics. As students could identify in more than one category, these percentages do not sum to 100%. In summary, the student population represented diverse institutions, backgrounds, and characteristics, underscoring applicability of the study findings across a broad range of engineering programs.

Table 1. Response proportion of the responses by university attended.

University	% of responses
Carnegie Mellon University	25.4%
University of Minnesota Duluth	19.5%
University of the Pacific	15.1%
Cal Poly Pomona	12.7%
New York University	9.5%
York College of Pennsylvania	9.5%
U.S. Coast Guard Academy	3.9%
University of Pittsburgh Johnstown	3.0%
University of Tennessee Knoxville	1.5%

Table 2. Response proportion of the responses by university funding type.

Category	% of responses
Public	59.5%
Private	40.5%

Table 3. Response proportion of the responses by respondent age.

Age	% of responses
Less than 21	42.3%
21 – 25	51.8%
Greater than 25	5.9%

Table 4. Response proportion of the responses by respondent grade point average (GPA).

GPA	% of responses
Less than 2.5	4.4%
2.6 - 3.0	13.9%
3.0 - 3.5	32.8%
Greater than 3.5	48.8%

Table 5. Response proportion of the responses by respondent year of study.

Year of study	% of responses
First Year	22.0%
Second Year	11.0%
Third Year	27.8%
Fourth Year	35.3%
Graduate	3.9%

Table 6. Response proportion of the responses by self-identified category. Note that students could select more than one category, so the responses do not sum to 100%.

Self-identified category	% of responses
Commuter	31.9%
First generation student	21.6%
Part time student	1.5%
Identify as an underrepresented individual	15.5%
Student athlete	14.6%
None of the Above	35.7%

Table 7. Response proportion of the responses by respondent gender.

Gender	% of responses
Man	55.7%
Woman	41.4%
Non-binary / non-conforming	1.2%
Prefer to self-describe	0.9%
Prefer not to say	0.6%
Transgender	0.3%

Summary of descriptor analysis

As described previously, 923 response items were characterized by 92 descriptors and classified into one of four categories, as shown in Appendix B. The frequency of occurrence of each descriptor was quantified. Twenty-five items that were cited 10 or more times are summarized in Table 8. These responses account for 76% of the descriptors. Supportive traits that relate to empathy, namely, “understanding”, “kind”, “caring”, were cited most frequently, followed by “helpful”. Motivational traits, namely, “passionate,” “fun/funny,” and “enthusiastic” were also considered important. “Knowledge/expertise” was the only professional trait ranked among the list shown. Among the results shown in Table 8, supportive traits account for 65%, motivational traits account for 24%, and personality and professional traits account for approximately 5% each.

The descriptors were analyzed for the draft of this paper (775 descriptors) and compared to responses after the additional 148 descriptors were included. In both cases, the seven descriptors shown at the top of Table 8 - “understanding” through “fun/funny” remained in the same position. The next fourteen descriptors - “enthusiastic” through “considerate” - remained among the top twenty one traits, although the order changed. The consistency in these observations supports the importance of these traits in building positive rapport.

Summary of response analysis

Figure 1 depicts a bar graph with mean and standard deviation of all faculty behavior items for all responses our survey received using the Likert scale from highest to the lowest. When professors “are respectful” received the highest equally-weighted (eq-weighted) average among all items listed, scoring 4.70 ± 0.07 (ranking #1). This was closely followed by when professors “use relevant class examples,” which had an average of 4.65 ± 0.07 (ranking #2), and “keep up with email” with 4.564 ± 0.07 (ranking #3). When professors “are enthusiastic” with an eq-weighted average of 4.58 ± 0.07 (ranking #4), and closely followed by when professors “show grace and understanding” with 4.58 ± 0.08 (ranking #5) (Figure 1).

On the other hand, among all respondents regardless of how the data were sliced, the least ranked item was “when professors stay late after class,” with an eq-weighted average of 3.48 ± 0.09 . In addition to this item, the bottom five behaviors also included when professors “arrive early to class”, “learn something about me”, “use inclusive language” and “reward comments and questions with praise” (Figure 1).

The overall analysis of the quantitative data showed that, when ranking all faculty behaviors by equally weighted means, the top five behaviors and the bottom five behaviors were clearly

identifiable across the entire group of respondents, regardless of demographic factors i.e. categorical variables in our study (Figure 2).

Table 8. Descriptors identified 10 or more times.

Trait	Frequency	Category	Lowman's linkage
Understanding	78	Supportive traits (IR)	IR
Kind	56	Supportive traits (IR)	IR
Caring	50	Supportive traits (IR)	IR
Helpful	48	Supportive traits (IR)	IR
Passionate	39	Motivational traits (IE)	IE
Knowledgeable/Expertise	37	Professional traits (IE)	IE
Fun/funny	36	Motivational traits (IE)	IE
Enthusiastic	31	Motivational traits (IE)	IE
Friendly	31	Supportive traits (IR)	IR
Encouraging	30	Supportive traits (IR)	IR
Engaging	27	Motivational traits (IE)	IE
Approachable	26	Supportive traits (IR)	IR
Supportive	26	Supportive traits (IR)	IR
Positive/happy	24	Supportive traits (IR)	IR
Patient	20	Supportive traits (IR)	IR
Respectful	20	Personality traits	PD/IR
Thoughtful	19	Supportive traits (IR)	IR
Honest	17	Personality traits	PD
Clear	15	Motivational traits (IE)	IE
Mentor/wants best for student	15	Supportive traits (IR)	IR
Considerate	13	Supportive traits (IR)	IR
Inspirational	11	Motivational traits (IE)	IE
Available	10	Supportive traits (IR)	IR
Empathetic	10	Supportive traits (IR)	IR
Organized	10	Motivational traits (IE)	IE

The data from responses to the Likert-scale questions were analyzed against six categorical variables, including institution, year of study, gender identity, age, GPA, and personal descriptors as described above. The analyses presented below are twofold: first, we report the top three behaviors ranked by respondents sliced by each categorical variable; second, we examine the data through high variance analysis. The top three substantial variations within groups for each categorical variable revealed the preferences most favored by each sub-demographic group. This is shown in Table 9.

By Institution:

When data is compared by institution, “use relevant class examples” had the highest equally weighted average of 4.70 ± 0.21 , followed by “are respectful” 4.67 ± 0.17 , and “keep up with email” 4.66 ± 0.21 . On the other hand, the least ranked item was “when professors stay late after class,” with an eq-weighted average of 3.51 ± 0.38 . When analyzing the responses by institution

depending on the high variance, three highest standard deviation values were observed for items including “stay late after class”, “use inclusive language” and “share their professional history and experience”. Respondents from the University of Tennessee Knoxville indicated a higher preference for their professors to stay late after class (eq weighted average of 4.20). Students from New York University indicated the strongest agreement towards “use inclusive language” (eq weighted average of 4.16).

Additionally, the item "when professors share their professional history and experience" was highly ranked by students from the University of Tennessee, Knoxville, with 100% of them strongly agreeing.

By year of study

When data is analyzed by the year of study, the top three behaviors were when professors “are respectful” with 4.63 ± 0.11 , “use relevant class examples” with 4.61 ± 0.16 , followed by “keep up with email” with 4.58 ± 0.11 . These behaviors consistently mattered a lot to the students.

When data is analyzed by the year of study according to the high variance, fourth-year students seemed to care the most about "when professors stay late after class" compared to other years (± 0.29), and graduate students seemed to care the least about this behavior indicating a mean score of 2.83. When professors “share their professional history and experience” matters most to the graduate students with a mean score of 4.42 (± 0.31). This closer look into the data revealed that third year students care most about when professors “arrive early to class” as compared to other years of study (± 0.26).

By gender identity

When considering gender identity, when professors “use relevant class examples” ranked as the top behavior with an eq-weighted average of 4.36 ± 0.94 , followed by “are enthusiastic” with 4.32 ± 0.85 , and followed by "keep up with email" at 4.28 ± 0.86 . The lowest ranked item was when professors “stay late after class," with an eq-weighted average of 2.82 ± 0.86 .

High variance analysis indicated that "when professors arrive early to class" (± 1.22) mattered most to men, with an eq-weighted average of 5.00, and least to individuals who “prefer to self describe/woman”. When professors "are respectful" (± 1.16) mattered most to Transgender/ non-binary/ non-conforming students, self describing woman, and woman transgender individuals, all groups with an eq-weighted average of 5.0. "When professors use inclusive language" (± 1.40) mattered most to “prefer to self describe/woman” and woman transgender individuals, with an eq-weighted average of 5.0.

By age

When data was analyzed by age, the highest-ranking behavior remained consistent, with when professors “are respectful" receiving an eq-weighted average of 4.63 ± 0.05 . The second highest-ranked behavior was "using relevant class examples," with an average of 4.56 ± 0.06 . "When professors keep up with email" was also among the top three behaviors, with an eq-weighted average of 4.54 ± 0.08 . Notably, "It is important to me that professors show grace or understanding" was highly valued, with an average of 4.51 ± 0.09 .

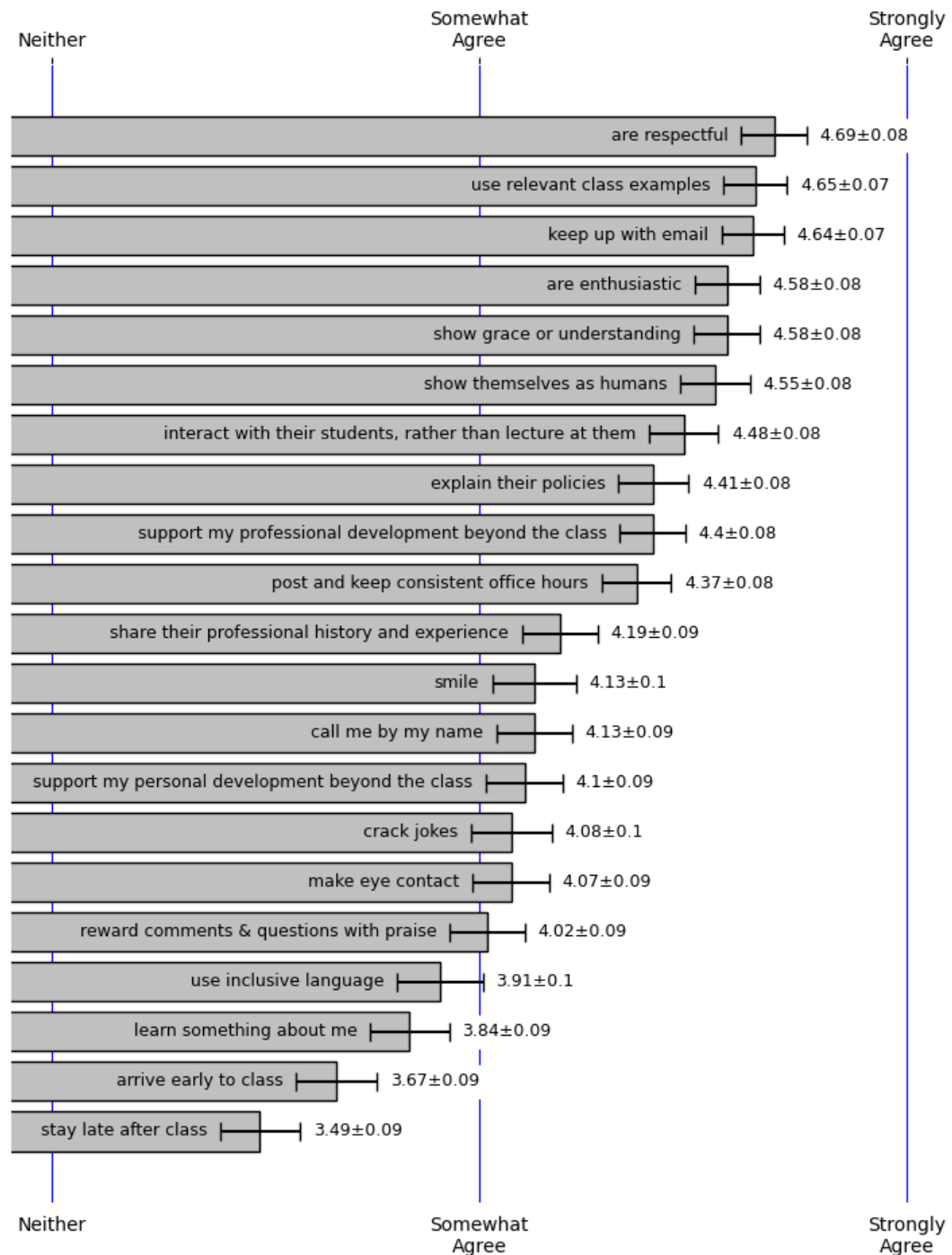


Figure 1: The ranking of importance for behaviors that establish positive rapport (highest ranking when professors are “respectful” and lowest “stay late after class”). Reflected is the mean as the gray bars, and the standard deviation as black whiskers, and labeled as mean $\pm$ standard deviation.

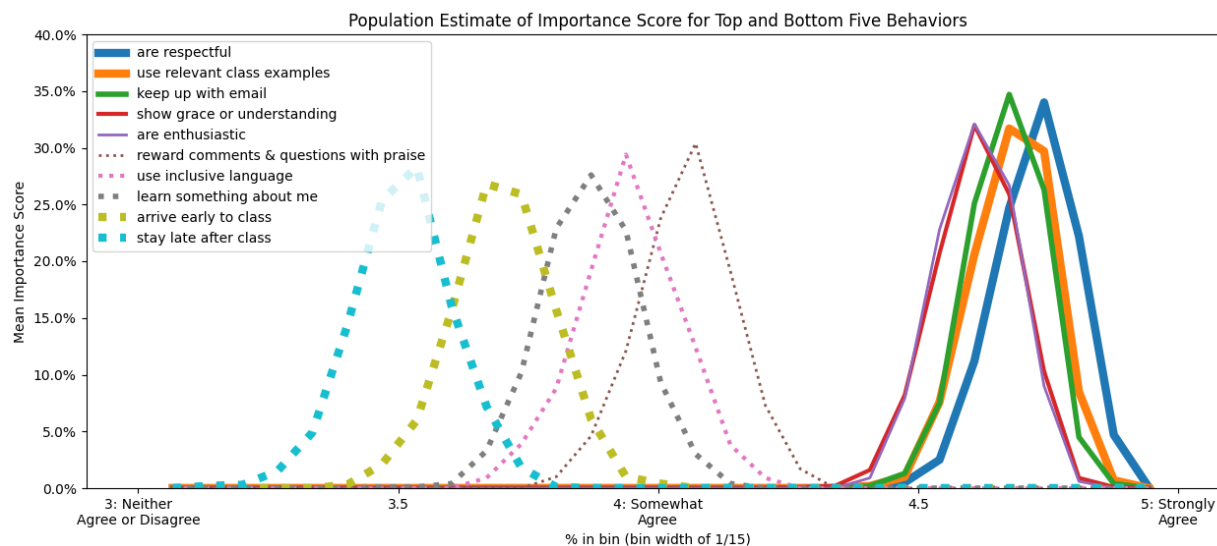


Figure 2: Population Estimate of Importance Score for Top and Bottom Five Behaviors. Shown here are the distributions (in bin sizes of 0.067 on importance score) of the population estimates resultant of the bootstrap analysis. One can see that the top five behaviors have a very similar distribution and mean, while the bottom five behaviors exhibit significantly more differentiation between the distribution estimates. No “Somewhat Disagree” or “Strongly Disagree” means were found in the bootstrapping process.

Variance analysis revealed that "learn something about me" had a significant standard deviation of 0.19, indicating that this behavior mattered more to students over 25. As students' age increased, the importance of "when professors learn something about me" and "share their professional history and experience" (± 0.25) became more significant for establishing positive rapport. When professors “arrive early to class” with (± 0.15) seemed to be highly valued by students between ages 21-25 with a mean score of 3.70.

By GPA:

When analyzing data by GPA, when professors “are respectful” was again the highest-ranked behavior, with an eq-weighted average of 4.62 ± 0.02 . This was followed by when professors “keep up with email” with 4.55 ± 0.12 and “show grace and understanding,” with an average of 4.52 ± 0.12 .

High variation was observed for when professors “arrive early to class” (± 0.32), “stay late after class” (± 0.20) and “share their professional history and experience” (± 0.28) were all mattered most consistently to students who reported a GPA > 2.5 and < 3.0 .

By Descriptors:

When considering personal descriptors, when professors “are respectful” was again the highest-ranked behavior, with an eq-weighted average of 4.78 ± 0.13 . This was followed by “when professors use relevant class examples” with a 4.72 ± 0.06 and “when professors keep up with email” at 4.69 ± 0.08 .

Variance analysis revealed that “rewarding comments and questions with praise” (± 0.36) was most significant to part-time students, with an average rating of 4.80. Additionally, “professors making eye contact” (± 0.27) was highly valued by part-time students, with an average rating of

4.60. Finally, "professors sharing their professional history and experience" (± 0.23) was also crucial for part-time students, receiving an average rating of 4.80.

Discussion

The first objective of this study was to identify faculty behaviors that have a higher potential for positive influence on rapport. From the analysis of responses to the open-ended question and the Likert-scale questions, it is clear that positive interpersonal behaviors such as respectfulness, understanding, kindness, and caring; professional behaviors such as answering emails in a timely fashion; and beneficial pedagogical approaches such as conveying enthusiasm for the subject and providing relevant examples in class are important for developing and maintaining positive rapport with students. Such supportive or motivational traits can be demonstrated in a variety of ways that are authentic to each faculty member's own personality. Some examples are provided below, from the authors' own experiences:

- Approachable: Tell stories about your own struggles and/or failures as an undergraduate student serves both to humanize you in the eyes of your students and provide a concrete, memorable, and cautionary tale to help them avoid common mistakes.
- Caring: Incorporate flexibility into the class policy as a way for the students to have flexibility on any given day, with minimal or no impact on their grade and learning experience. For example, distribute two or three passes for late submittals at the start of the semester with clear guidance as to how to use them. The intent of the late pass is not for emergencies but instead for days when there was a personal, social, or other event when students may need to modify their schedule. The pass gives students some flexibility, but holds them accountable for completing their work.
- Caring: Conduct a survey before your course begins to identify students' interests. Returning to the survey results throughout the semester and asking students questions about their interests helps to establish a supportive atmosphere.
- Clear: Include different statements in your syllabi to guide the class on building an enjoyable and productive learning environment. An inclusion or inclusive statement (everyone's voice is important and we are a team), a statement of what is expected from the instructor (instructor enthusiastically will guide the ship and support everyone until the end of the semester), and a statement of what is expected from each student (get ready to practice a lot and to challenge yourself; you are not alone; little steps will take you really far; be present and try your best; and if something doesn't work be ready to try it again) are building blocks to developing an enjoyable and productive learning environment.
- Considerate: Use "please" and "thank you" in oral and written communication to model professional behavior and demonstrate your respect for students
- Encouraging: Have an anonymous interactive check-in activity at the start of each week, during exam season, or at any time that the instructor perceives a change in the energy in the classroom dynamics, will allow instructors to learn how students are feeling when they are coming to class. This outreach sends the message that the professor cares about the students. Depending on the situation, the professor can provide campus resources available to support students' wellbeing or if any student has identified themselves by asking for further support, the professor can direct the student to a specific resource.

Table 9. Mean Importance Score of "It's important to me that my professors" Questions ($\mu \pm \sigma$). The scores of each question were calculated though bootstrapping. Questions where the mean is with in the top three for that category, indicating high importance, are indicated in bold-underline. Those where the standard deviation is within the top three, indicating a diverse range in responses between labels in that category, are indicated in bold-italic.

Questions	Overall	Institution	Study Year	Gender	Age	GPA	Desc
call me by my name	4.12 $\pm$ 0.09	4.07 $\pm$ 0.25	4.03 $\pm$ 0.19	3.72 $\pm$ 1.04	4.05 $\pm$ 0.09	4.04 $\pm$ 0.08	4.15 $\pm$ 0.21
learn something about me	3.83 $\pm$ 0.09	3.79 $\pm$ 0.31	3.76 $\pm$ 0.22	3.37 $\pm$ 0.51	<u>3.83$\pm$0.19</u>	3.77 $\pm$ 0.10	3.86 $\pm$ 0.05
use relevant class examples	<u>4.65$\pm$0.08</u>	<u>4.70$\pm$0.21</u>	<u>4.61$\pm$0.16</u>	<u>4.36$\pm$0.94</u>	<u>4.56$\pm$0.06</u>	4.51 $\pm$ 0.15	<u>4.72$\pm$0.06</u>
arrive early to class	3.67 $\pm$ 0.09	3.61 $\pm$ 0.21	<u>3.49$\pm$0.26</u>	<u>3.23$\pm$1.22</u>	<u>3.55$\pm$0.15</u>	<u>3.52$\pm$0.32</u>	3.75 $\pm$ 0.15
stay late after class	3.49 $\pm$ 0.09	3.51 $\pm$ 0.38	<u>3.30$\pm$0.29</u>	<u>2.82$\pm$0.89</u>	3.38 $\pm$ 0.12	<u>3.45$\pm$0.20</u>	3.38 $\pm$ 0.17
explain their policies	4.41 $\pm$ 0.09	4.35 $\pm$ 0.19	4.36 $\pm$ 0.12	4.08 $\pm$ 0.84	4.34 $\pm$ 0.03	4.28 $\pm$ 0.11	4.48 $\pm$ 0.19
post and keep consistent office hours	4.36 $\pm$ 0.08	4.30 $\pm$ 0.28	4.34 $\pm$ 0.14	4.06 $\pm$ 1.13	4.25 $\pm$ 0.14	4.34 $\pm$ 0.07	4.42 $\pm$ 0.15
keep up with email	<u>4.64$\pm$0.07</u>	<u>4.66$\pm$0.21</u>	<u>4.58$\pm$0.11</u>	<u>4.28$\pm$0.86</u>	<u>4.54$\pm$0.08</u>	<u>4.55$\pm$0.12</u>	<u>4.69$\pm$0.08</u>
interact with students, rather than lecture	4.47 $\pm$ 0.08	4.41 $\pm$ 0.19	4.37 $\pm$ 0.11	4.10 $\pm$ 1.01	4.39 $\pm$ 0.09	4.43 $\pm$ 0.09	4.55 $\pm$ 0.19
reward comments/questions with praise	$\pm$ 4.02 $\pm$ 0.09	3.97 $\pm$ 0.31	3.96 $\pm$ 0.18	3.98 $\pm$ 0.89	4.00 $\pm$ 0.14	3.96 $\pm$ 0.07	<u>4.19$\pm$0.36</u>
are enthusiastic	4.58 $\pm$ 0.07	4.58 $\pm$ 0.15	4.49 $\pm$ 0.08	<u>4.32$\pm$0.85</u>	4.48 $\pm$ 0.09	4.50 $\pm$ 0.10	4.66 $\pm$ 0.22
crack jokes	4.07 $\pm$ 0.09	4.11 $\pm$ 0.28	3.95 $\pm$ 0.17	3.61 $\pm$ 0.79	3.96 $\pm$ 0.10	4.01 $\pm$ 0.18	4.12 $\pm$ 0.09
make eye contact	4.07 $\pm$ 0.09	4.08 $\pm$ 0.24	4.01 $\pm$ 0.14	3.37 $\pm$ 0.87	4.05 $\pm$ 0.10	4.06 $\pm$ 0.10	<u>4.18$\pm$0.28</u>
are respectful	<u>4.69$\pm$0.07</u>	<u>4.67$\pm$0.17</u>	<u>4.63$\pm$0.11</u>	<u>4.07$\pm$1.16</u>	<u>4.63$\pm$0.05</u>	<u>4.62$\pm$0.02</u>	<u>4.78$\pm$0.13</u>
smile	4.13 $\pm$ 0.10	4.11 $\pm$ 0.22	3.95 $\pm$ 0.21	3.52 $\pm$ 1.13	4.00 $\pm$ 0.12	4.05 $\pm$ 0.12	4.13 $\pm$ 0.17
use inclusive language	3.91 $\pm$ 0.10	3.74 $\pm$ 0.38	3.78 $\pm$ 0.14	<u>3.49$\pm$1.40</u>	3.88 $\pm$ 0.11	3.83 $\pm$ 0.14	3.96 $\pm$ 0.21
show grace or understanding	4.58 $\pm$ 0.08	4.51 $\pm$ 0.15	4.49 $\pm$ 0.14	4.04 $\pm$ 1.15	4.51 $\pm$ 0.09	<u>4.52$\pm$0.12</u>	4.67 $\pm$ 0.21
support professional development beyond class	4.41 $\pm$ 0.08	4.38 $\pm$ 0.16	4.32 $\pm$ 0.13	4.00 $\pm$ 0.91	4.31 $\pm$ 0.12	4.37 $\pm$ 0.07	4.49 $\pm$ 0.07
support personal development beyond class	4.10 $\pm$ 0.09	4.02 $\pm$ 0.20	3.96 $\pm$ 0.17	3.48 $\pm$ 1.13	3.97 $\pm$ 0.12	4.07 $\pm$ 0.18	4.11 $\pm$ 0.07
share their professional history and experience	4.19 $\pm$ 0.09	4.27 $\pm$ 0.33	<u>4.14$\pm$0.31</u>	3.77 $\pm$ 0.78	<u>4.19$\pm$0.25</u>	<u>4.06$\pm$0.28</u>	<u>4.41$\pm$0.23</u>
show themselves as humans	4.55 $\pm$ 0.08	4.57 $\pm$ 0.20	4.51 $\pm$ 0.17	4.16 $\pm$ 0.94	4.50 $\pm$ 0.01	4.46 $\pm$ 0.12	4.65 $\pm$ 0.21

- Kindness: Give students a stretch break in the middle of class so they can catch up, stretch, check their phones, talk to a neighbor, or just take a break, then return their focus to learning once the break is over [28].
- Knowledgeable/Expertise: Use relevant current or breaking news in the classroom to help students build connections of course topics to real life examples. This practice can be enhanced further by showing students how the skills acquired in the classroom can be used to improve or support the community's education and development, right away or once they graduate. For example, breaking news about a disaster-flooding, fires, etc- that has caused immense damage can be used to help students find focused motivation and energy to concentrate on their learning, and keep in mind the importance of their new skills that are applicable to making a difference in their communities.
- Passionate: Convey enthusiasm for the subject and for the students by greeting the students, playing music before class, bringing relevant demonstrations or videos, sharing your own personal knowledge and experiences, and encouraging students. Conveying passion and enthusiasm engages students in the class.
- Respectful: Maintain consistent office hours, listen to students' questions, and treat students as "junior engineers." These are examples of respectful behavior that help to establish expectations of a professional environment in and out of the classroom.
- Supportive and Encouraging: Avoid basing most of the course grade on high-stakes deliverables such as exams. Grading should be broken down into different types of experiences, including some that allow students to learn from mistakes. For example, allowing students to rewrite papers, or redo practice problems, etc., and regrading the work, increases students' grades while giving them an opportunity to reflect on instructor's feedback and review technical material needed to get a higher grade.
- Understanding: Adjust a deadline to accommodate students who have multiple assignments due at the same time. Allow students to earn extra credit for submitting the work by the original deadline. Students who need the deadline extension have more time to complete the assignment, but those who complete the work on time are recognized/rewarded for on-time submission.

The analysis of the Likert-scale responses across various categorical variables reveals a consistent pattern in student preferences for professor behaviors (Figure 3). Notably, "when professors are respectful" emerged as the most impactful behavior for establishing positive rapport, regardless of the category analyzed. This finding underscores the universal importance of respect in the student-professor relationship.

The consistency of the top five behaviors across all categories highlights key aspects that students value in their interactions with professors (Figure 3). These behaviors include being respectful, using relevant class examples, keeping up with email, demonstrating enthusiasm, and showing grace or understanding. The emphasis on these behaviors suggests that students appreciate professors who are not only professional and organized but also approachable and empathetic. When professors "are respectful" was the top ranked behavior four out of the six categorical variables namely by year of study, Age, GPA, and descriptors.

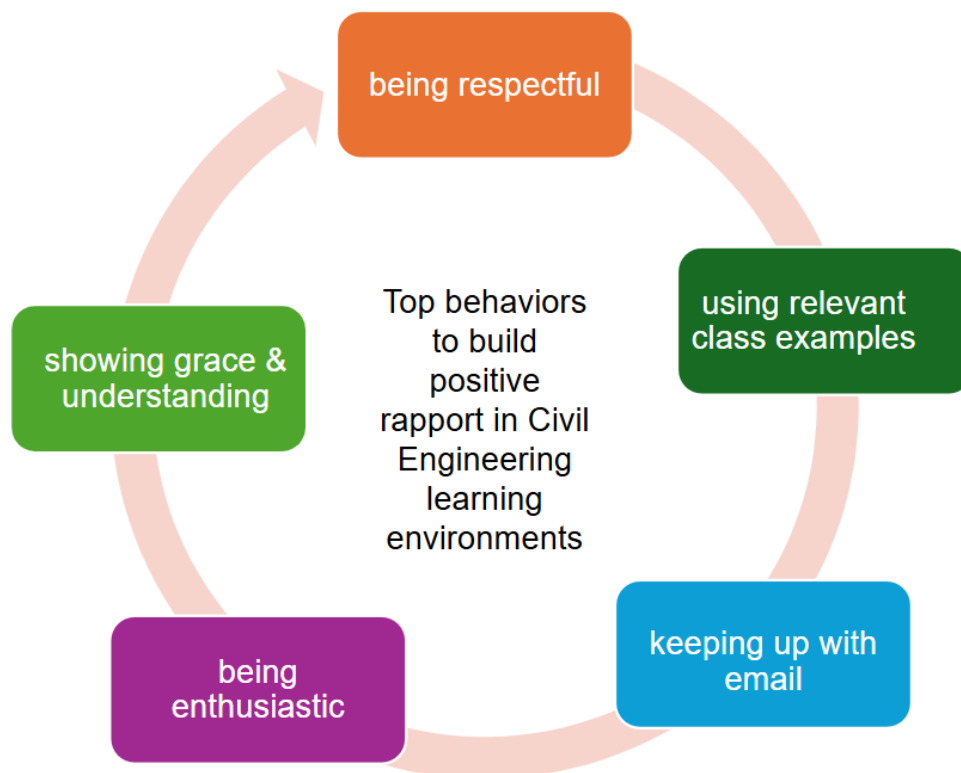


Figure 3: Top rated behaviors exhibited by civil engineering professors to establish positive rapport with students.

Interestingly, the importance of being respectful and using inclusive language was particularly pronounced when analyzed by gender identity, indicating that students from diverse gender backgrounds place a higher value on inclusivity and respect. While students over 25 years of age valued professors learning something about them and sharing their professional history more, students aged 21-25 highly valued professors arriving early to class. While upper level undergraduate students, and students with a GPA between 2.5 and 3.0 highly valued professors staying late after class or arriving early to class, graduate students prioritize professors sharing their professional history. Students with a GPA between 2.5 and 3.0 consistently showed high variations for their preferences for faculty behaviors. Among the groups who showed variation towards some faculty behaviors were also part-time students, this non-traditional student group highly valued when professors are rewarding comments with praise, making eye contact, and sharing their professional history as compared to other students.

Behaviors such as staying late after class, arriving early, and learning something personal about the students consistently ranked lower than others. This finding suggests that while students value certain professional and interpersonal behaviors, they may not prioritize additional time commitments or personal engagement to the same extent.

Conclusion

As part of a course and lesson design, faculty should consider rapport-building strategies as an essential part of the process. Such strategies should be authentic to the instructor but they should consider the needs and perspectives of the learners. Strategies used to establish positive rapport should be intentionally selected and their effectiveness should be assessed by asking for students' feedback.

Results of this study highlight useful and effective rapport-building behaviors that professors can adopt, emphasizing the universal importance of respect and understanding. Demographics such as gender identity, age, and year of study influence students' opinions of rapport-building actions and behaviors. However, regardless of student differences, the core five behaviors, being respectful, using relevant class examples, keeping up with email, being enthusiastic, and showing grace or understanding remain crucial for fostering a supportive and engaging learning environment in today's higher education classrooms.

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Appendix A: Survey Introduction and Questions

You are being invited to participate in this multi-institutional research study on the impact of faculty behaviors on student-faculty rapport.* The purpose of this study is to obtain some information about students' perceptions of their relationship with professors. We hope to use this information to improve the relationship between professors and engineering students. In addition, we will share the aggregate results in presentations and/or publications. This questionnaire will take approximately ten minutes to complete. This survey is anonymous. Please give us your honest opinion. Participating in this survey is voluntary. We greatly appreciate your assistance with this study. The York College of Pennsylvania IRB has reviewed this research project and found it to be exempt under Exemption 2 (45r CFR 46.401(d)). If you have questions or concerns, please contact the authors. Thank you!

*Merriam-Webster defines "rapport" as: "a relationship characterized by agreement, mutual understanding, or empathy that makes communication possible or easy." (<https://www.merriam-webster.com/dictionary/rapport>) ("Rapport Definition & Meaning", n.d.).

Consent: By submitting this survey, I affirm that I am completing this survey, for the first time, to the best of my understanding. I also agree that the information may be used in the research project described above. Please indicate Yes or No.

Think of a professor(s) who has or had a positive influence in your life or education. Provide (3) words or phrases that describe this person's interaction with you, or your observations of their interaction with others, that left an impression on you.

It's important to me that my professors ... (rate each strongly disagree, somewhat disagree, neither agree nor disagree, somewhat agree, or strongly agree)

Call me by my name.

Learn something about me.

Use relevant class examples.

Arrive early to class.

Stay late after class.

Explain their policies.

Post and keep consistent office hours.

Keep up with email.

Interact with their students, rather than lecture at them.

Reward comments/questions with praise.

Are enthusiastic.

Crack jokes.

Make eye contact.

Are respectful.

Smile.

Use inclusive language.

Show grace or understanding.

Support my professional development beyond the class.

Support my personal development beyond the class.

Share their professional history and experience.

Show themselves as humans.

Please select your institution.

Please indicate your year of study.

Please indicate your gender identity.

Please indicate your age.

Please indicate your overall GPA.

Please choose any options that describe you. (Commuter (i.e. do not live on or near campus), Identify as an underrepresented individual, First generation student, Part time student, Student athlete)

Appendix B: Descriptor Summary: Descriptors classified by category

Personality Traits	Supportive Traits (IR)	Motivational Traits (IE)	Professional Traits (IE)
Adaptable	Accommodating	Challenging	Creative
Appreciated	Approachable	Clear	Demos / examples
Calm	Attentive	Command of the	Effective
Charming	Available	classroom	Knowledgeable /
Confident	Caring	Communication	expertise
Dedicated	Compassionate	Curious	Mastery grading
Driven	Connection	Demanding	Practical
Eccentric	Considerate	Energetic	Professional
Focused	Constructive	Engaging	Rigorous
Genuine	Empathetic	Enthusiastic	
Grounded	Encouraging	Firm	
Gruff	Fair	Fun / funny	
Honest	Feedback	High expectations	
Humble	Flexible	Informative	
Respectful	Friendly	Insightful	
Sensible	Generous	Inspirational	
Stern / strict	Gives advice	Intentional	
Strong	Go above and beyond	Interactive	
Wise	Helpful	Interesting	
	Inclusive	Leader	
	Invested	Motivating	
	Kind	Organized	
	Listens	Outgoing	
	Mentor / wants best	Passionate	
	for student	Thorough	
	Nice		
	Nonjudgmental		
	Open		
	Patient		
	Perceptive		
	Personable		
	Personal		
	Positive / happy		
	Present		
	Receptive		
	Relatable		
	Resourceful		
	Responsive		
	Supportive		
	Thoughtful		
	Understanding		
	Welcoming		